Highlights

WaveSplat: Grid-Free Wave Field Estimation from Sparse 3D Point Clouds
with a Learned Spectral Prior

Saeyong Park

- Grid-free wave estimation: continuous-wavenumber atoms with a learned prior.
- Beats the strongest eligible grid baseline on crowded synthetic seas.
- Survives a non-linear reference field and seeds outside the training corpus.
- Ablations attribute the gain to the prior and to wavenumber continuity.
- On measured records it does not reach the pre-registered decision threshold.

WaveSplat: Grid-Free Wave Field Estimation from Sparse 3D Point Clouds with a Learned Spectral Prior

Saeyong Park[a,*]

[a] *Avikus (HD Hyundai), Republic of Korea*

Abstract

Phase-resolved ocean wave fields must be reconstructed from point clouds that are sparse and irregular. The prevailing approach assimilates such returns into a spatial grid state, which requires interpolating them onto that grid; the closest recent study measures direct assimilation of the sparse returns and reports limited improvement at their original locations, and its outlook names a spectral-state formulation. This paper is that formulation.

We represent the field as a small set of continuous-wavenumber spectral atoms constrained by the dispersion relation, place them with a learned set-valued prior, and refine them by variable projection before updating per frame. We claim no novelty for sequential or real-time operation, which this literature already has; the contribution is the combination, and an intervention-based account of what each part of it buys.

On crowded multi-component seas the estimator lowers median normalised RMSE by -38.6% against the strongest eligible grid baseline we possess (62/64 episodes, synthetic). The result survives a non-linear reference field with seeds outside the training corpus (43/48), a case in which the baseline itself retains skill (47/48), and a physically correct range-dependent sensor model. Crucially the claim is carried by ablation rather than by baseline comparison alone: removing the learned prior costs -36.3%, and removing only the continuity of the

*Corresponding author
Email address: saeyong.park@avikus.ai (Saeyong Park)

wavenumber costs -38.6% with a dose response. The gain also persists inside the observation-density band our measured records occupy and at a forecast rung most of them cover — though in the latter the comparator degrades over the same interval, so the ratio grows while the accuracy does not. The two ablation magnitudes are not additive and are not shares of the headline: different corpora, different comparison arms, and we checked the wavenumber ablation separately for confounding with the refit it performs.

Scope. These gains are synthetic and confined to the crowded stratum; the sparse stratum does not separate. On 12 fixed-station stereo records the learned prior does not reach the pre-registered decision threshold against a per-record zero-field reference under any configuration we tested. We report that negative, and we report the limits of the procedure that produced the positives: the ablation comparators are themselves weak against a zero predictor, the checkpoint was trained on the same distribution family as the synthetic evaluation, and the re-scoring studies were not blind. Section 5.6 states each in full.

Keywords: wave field estimation, LiDAR, spectral methods, dispersion relation, data assimilation, real-time estimation

---

1. Introduction

1.1. Motivation: wave field information for maritime autonomy

Table 1 summarizes the notation used throughout. Knowledge of the surrounding ocean wave field is a prerequisite for many functions in autonomous and semi-autonomous ship operations [1, 2]: short-term prediction for proactive motion compensation during cargo handling and helicopter landing [3], route optimization to minimize fuel consumption and fatigue loading [4], wave-preview feedforward ride control on high-speed hydrofoil vessels [5], and structural health monitoring [6]. In each application the value of the estimate depends on its spatial resolution, update rate, and accuracy.

Table 1: Nomenclature.

Symbol	Description
$z(x, y, t)$	Surface elevation [m]
N	Number of spectral components
w_i	Amplitude of i-th component [m]
$k_{x,i}, k_{y,i}$	Wavenumber components [rad/m]
ω_i	Angular frequency [rad/s]
φ_i	Phase offset [rad]
g	Gravitational acceleration [m/s^2]
d	Water depth [m]
H_s	Significant wave height [m]
T_p	Peak period [s]
c_j	Confidence weight for observation j
θ_j	Beam incidence angle [rad]
r_j	Slant range to observation j [m]
M	Number of LiDAR observations per frame
nRMSE	RMSE normalized by H_s

1.2. LiDAR point clouds for ocean surface sensing

Among sensor modalities, wave buoys [7] lack spatial resolution, X-band marine radar [8, 9] images large areas but measures backscatter rather than elevation, and satellite altimetry [10] is too coarse for vessel-scale real-time use. Scanning LiDAR, by contrast, provides *direct* elevation measurements over a wide area at ∼10 Hz [11]. That capability is not hypothetical: airborne scanning lidar has resolved open-ocean wave fields and their directional wavenumber spectra since the 1990s [12], from aircraft rather than from ships and at a correspondingly larger footprint. We note this explicitly because the absence of real-data validation in the present work (Section 6.6) reflects the absence of a *publicly archived* phase-resolved lidar sea-surface dataset, not the absence of

such measurements. Current automotive-grade solid-state LiDAR systems operating at eye-safe wavelengths (1550 nm) offer >100 channels, wide fields of view ($\sim$120° × 25°), and ranges up to 250 m with centimeter-level accuracy in the near field. However, LiDAR observations are sparse (50–200 valid returns per scan), unevenly distributed (denser near the ship), and noisy (degrading with range and incidence angle). The challenge is therefore to reconstruct a continuous, physically consistent wave field from these sparse, heterogeneous observations in real time.

1.3. Grid-based paradigm and its limitations

The dominant approach follows a grid-based data assimilation paradigm adapted from numerical weather prediction: discretize space into a regular grid, represent the wave state at grid points, and apply an ensemble Kalman filter (EnKF/ETKF) with FFT-based spectral forecasting [13, 3, 14]. While principled, this approach has three limitations under sparse LiDAR: (i) *interpolation artifacts*—sparse returns must be mapped to grid points, introducing smoothing near the Nyquist limit (5 m grid → $\sim$10 m minimum wavelength regardless of LiDAR density); (ii) *fixed resolution* independent of scene complexity, wasting capacity on simple seas and under-resolving complex ones; and (iii) *forecast cost* $O(N_e G^2 \log G^2)$ per step regardless of true wave complexity, e.g. 50 independent 2D FFTs for N_e=50, G=40×30.

1.4. Proposed approach

We propose **WaveSplat**[1], a grid-free spectral representation that eliminates the spatial grid entirely. Instead of discretizing space, we represent the wave field as a weighted superposition of N plane waves:

$$z(x, y, t) = \sum_{i=1}^{N} w_i \cos(k_{x,i} x + k_{y,i} y - \omega_i t + \varphi_i) \tag{1}$$

[1]The name reflects "wave" + "splat" in the sense of projecting (splatting) spectral components onto arbitrary spatial locations, analogous to point-based rendering; it does not imply a methodological connection to 3D Gaussian Splatting [15].

where each spectral component is parameterized by amplitude w_i, wavenumber $(k_{x,i}, k_{y,i})$, and phase φ_i. The angular frequency ω_i is not a free parameter but is *constrained* by the linear dispersion relation:

$$\omega_i = \sqrt{g\,|k_i|\,\tanh(|k_i|\,d)} \tag{2}$$

where $g = 9.81$ m/s^2 and d is the water depth. This hard constraint reduces the parameter space from $5N$ to $4N$, regularizes optimization, and ensures physically plausible wave propagation during temporal tracking.

Our key contributions are:

1. **Grid-free spectral estimation framework (C1)**: Direct estimation on sparse, irregularly distributed observations without grid interpolation, enabling wave field evaluation at arbitrary spatial resolution. Unlike single-snapshot sparse spectral methods [16], the spectral component representation persists across frames, allowing computationally efficient per-frame updates via dispersion-based phase prediction ($\sim$10 gradient steps) rather than full re-initialization ($\sim$200 steps per frame), while supporting continuous monitoring without the discrete update latency of filter-based approaches. Estimation accuracy depends on initialization quality; tracking refines amplitudes and phases given the wavenumbers established during initialization.
2. **Dispersion-constrained parameterization (C2)**: Hard enforcement of the full dispersion relation (including tanh depth dependence) reduces the free parameter count and acts as a physics-based regularizer, improving tracking stability over extended time horizons. *[SCOPE 2026-08-14]* The claim made here is the parameter-count reduction. Whether this constraint contributes to *accuracy* **has not yet been measured**: the ablation arm previously used to support it removed the frequency initialisation together with the constraint, and a controlled intervention that isolates the constraint was null in both strata.
3. **Geometry-aware confidence weighting (C3) — [RETRACTED 2026-08-14]**: Per-observation weights $c_j = \cos\theta_j/(r_j/d_{\text{ref}})$ based on inci-

dence angle and range were introduced to account for the heteroscedastic noise structure of LiDAR measurements on ocean surfaces. This is **withdrawn as a contribution**: on a fresh out-of-sample corpus, disabling the weighting was *better* in both the sparse and the crowded stratum. It is retained below only as a reported negative result.

The pipeline also includes an online model-order mechanism (energy-based pruning and residual-triggered growth, the retracted corpus (archived)) that matches the number of spectral components to scene complexity, from $N{=}2$ for simple swells to $N{=}21$ for complex multi-component seas. We deliberately do *not* list it as a contribution: measured under the corrected tracker with a median-of-3 seed protocol, its accuracy benefit is null (retracted corpus; archived), and we discuss it as a mechanism and a compute-efficiency device in Section 6.6 instead.

Relative to prediction-error frameworks for deterministic wave forecasting such as LEPrE [17], which quantify the error of a linear-fitting prediction procedure without proposing an estimator, we contribute the estimator itself: a nonlinear, dispersion-constrained, grid-free fit with online model order, evaluated on scanning-LiDAR sampling geometry.

The remainder of this paper is organized as follows. Section 2 reviews related work on wave field estimation. Section 3 presents the WaveSplat algorithm in detail. Section 4 describes the simulation environment and evaluation metrics. Section 5 presents experimental results including baseline comparisons, ablation studies, and robustness analysis. Section 6 discusses the findings and limitations. Section 7 concludes.

2. Related Work

We organize the review around two axes: (1) the sensor modality used for ocean surface observation, and (2) the estimation algorithm used to reconstruct the wave field from observations. Table 2 provides a quantitative comparison of sensor modalities.

Table 2: Comparison of ocean surface sensing modalities for vessel-scale wave field estimation. Typical values for operational systems.

Sensor	Spatial range	Resolution	Update rate	Direct elevation?	Key limitation
Wave buoy	Point	—	1 Hz	Yes	No spatial information
X-band radar	0.5–4 km	∼7.5 m	1–2 s/rev	No (intensity)	Requires SNR inversion
Satellite (altimeter)	Global	∼7 km	Days	Yes (nadir)	Coarse, non-real-time
Shipboard LiDAR	10–250 m	∼0.1 m (near)	10 Hz	Yes	Sparse, range-dependent noise
Ship motion (RAO)	—	—	Continuous	No (indirect)	Limited to encountered waves

2.1. Conventional shipboard wave sensing

X-band radar is the most extensively studied sensor for shipboard wave estimation. Young et al. [9] first demonstrated that radar backscatter modulation (tilt, hydrodynamic, shadowing) carries wave spectral information, and Nieto-Borge et al. [8] recovered the directional wave spectrum via 3D FFT in the (k_x, k_y, ω) domain with the linear dispersion relation serving as a spectral filter. This "dispersion filtering" concept is conceptually related to our dispersion-constrained parameterization, though applied in the Fourier domain rather than as a hard parameter constraint. Radar's fundamental limitation is that it measures backscatter intensity rather than surface elevation directly, requiring an empirical modulation transfer function that depends on radar parameters, wind speed, and sea state [8]; its spatial resolution (∼7.5 m) is also coarser than LiDAR (∼0.1 m in the near field). **Ship-motion-based** methods instead invert the vessel RAO to recover the incident wave spectrum from heave/pitch/roll measurements [18, 19]; these require no additional sensors but cannot resolve the spatial wave field, only the directional spectrum at the vessel location.

2.2. Grid-based wave field reconstruction

Deterministic wave field reconstruction from spatially resolved LiDAR/radar has been dominated by spatial discretization combined with ensemble data assimilation. Desmars et al. [13] introduced ETKF-based reconstruction from X-band radar using spectral propagation on a regular grid; the Ensemble Transform Kalman Filter [20] is preferred over the stochastic EnKF [14] for its de-

terministic analysis step. Desmars et al. [3] extended this framework with a nonlinear reconstruction-and-prediction scheme on remotely measured surfaces, demonstrating short-term forecasting up to one peak period ahead. Adaptation of ensemble Kalman filtering to scanning shipboard LiDAR has, to our knowledge, not been reported in the open literature; we therefore implement an ETKF-grid baseline by porting the radar-based formulation of Desmars et al. [13] to LiDAR observations (retracted corpus; archived). These grid-based methods share a common architecture—discretize the 2D domain into a regular grid, maintain an ensemble of spectral states, and assimilate observations via the Kalman update—which introduces an inherent trade-off between spatial resolution and computational cost, with interpolation from sparse observations to grid points being a known source of error. We distinguish this *deterministic* reconstruction from the *stochastic* wave prediction performed by large-scale spectral models such as WAVEWATCH III [21], which forecast bulk statistics at kilometer scales without resolving individual phases.

A complementary line of work asks not how to reconstruct the field but how large the resulting prediction error is. Fucile et al. [17], in this journal, model the surface elevation as a Gaussian stochastic process and derive a semi-analytical Linear Estimator of Prediction Error (LEPrE), which returns the prediction-error standard deviation in time and space for a phase-resolved linear wave model whose coefficients are obtained by linear fitting on measurements, and which places the notion of a "Predictability Region" on a probabilistic footing; the formulation is verified against Monte Carlo simulation for long- and short-crested seas. Our contribution is orthogonal on three axes. First, in *what is produced*: Fucile et al. [17] characterise the error of a given deterministic prediction procedure and do not propose a field estimator, whereas we propose one and benchmark its reconstruction accuracy against ensemble-Kalman and Gaussian-process baselines. Second, in the *estimator class*: LEPrE presumes model coefficients determined by linear fitting, while WaveSplat obtains $(w_i, k_{x,i}, k_{y,i}, \varphi_i)$ by nonlinear joint optimisation with the dispersion relation imposed as a hard constraint and with the model order adapted online—a class outside the linear-

fitting assumption on which the LEPrE derivation rests. Third, in the *evidence base*: the LEPrE formulation is agnostic to the measurement device, whereas our evaluation is specific to the sampling geometry of a scanning shipboard LiDAR, and that geometry is *simulated*: the shipboard configuration in this paper exists only in the sensor model. The measured records used in Section 5.6 come from fixed stations and from a different sensor modality, so no result here is supported by measured shipboard data, including sparse, range- and incidence-dependent returns and reflectivity-driven dropout. The two are naturally combined rather than opposed: WaveSplat currently returns point estimates without an error model (Section 6.6), and the LEPrE framework is the natural route to supplying one for the grid-free spectral representation.

2.3. Position of this work relative to the closest grid-based study

The closest study to ours assimilates sparse three-dimensional returns into a *spatial grid* state with an ensemble transform Kalman filter [22]. Two things about it shape our position.

First, sequential and continuously-updating operation is *not* what separates this paper from that line: recursive per-frame filters already exist there, and we therefore make no novelty claim on that axis. What differs is the state itself — a small set of continuous-wavenumber atoms rather than a grid — and the fact that the atoms are placed by a learned prior.

Second, and more usefully, that study does not merely leave direct sparse assimilation as future work: it *measures* it, and reports limited improvement at the original sparse locations relative to interpolating the returns first. Our contribution is therefore better described as a spectral-state implementation of an intervention whose grid-state version has already been tried and found wanting, than as an untried idea. We are careful not to over-read the outlook of that paper: it points at a spectral-state formulation, not at a learned prior or at continuous wavenumbers, and the ablations in Section 5.4 are what carry those two components here.

2.4. Grid-free alternatives: spectral, GPR, and learning-based

Several alternatives to grid-based assimilation have been explored. Classical **spectral estimation** methods (FFT, Welch, MUSIC, ESPRIT) assume regular sampling [23]; Capon's minimum-variance beamforming [24] and MUSIC [25] handle irregular arrays but either provide only power spectra without phases (Capon) or require known model order (MUSIC). WaveSplat leverages beamforming for *initialization* and refines the estimates by gradient optimization with physics constraints, combining beamforming robustness with joint amplitude-phase recovery. **Gaussian process regression** [26] with Matérn or squared-exponential kernels provides nonparametric spatial interpolation but scales as $O(M^3)$ and, in its standard single-frame form, lacks a temporal propagation model; sparse approximations [27] mitigate cost but still treat the wave field as a generic spatial random process rather than a dispersion-constrained propagating signal. **Learning-based** methods include physics-informed neural networks [28, 29], the U-Net/Fourier neural operator of Ehlers et al. [30] for phase-resolved reconstruction, and the nonlinear ETKF extension of Desmars et al. [3]. These approaches show strong results on their training distributions but face practical challenges for arbitrary shipboard sensor configurations: FNO-style models assume uniform training grids incompatible with scanning-LiDAR irregular sampling, generalization across sensor geometries is limited, and the black-box nature complicates failure diagnosis. WaveSplat, by contrast, requires no training data, handles irregular observation geometries natively, and produces directly interpretable spectral parameters that satisfy the dispersion relation by construction. Direct experimental comparison with learning-based methods is beyond the scope of this work, which focuses on physics-constrained spectral estimation; such a comparison would require careful control of training distribution, sensor geometry, and computational budget. Its grid-free sparse representation is conceptually related to the compressed sensing literature [31, 32] and to Donelan et al. [16]'s directional spectral analysis of propagating waves, with three key differences: (1) each basis function is a physically meaningful plane wave with frequency constrained by the dispersion relation rather than a generic

sinusoid or wavelet; (2) the model order is adapted *online* during tracking based on residual energy, rather than fixed a priori; and (3) the spectral components *persist* across frames via dispersion-based phase prediction, enabling efficient per-frame updates ($\sim$10 gradient steps) rather than independent per-frame reconstruction, which is the core enabler of real-time operation.

3. Method

3.1. Forward model

The wave field is modeled as a superposition of N linear plane waves (1). Each component i has four free parameters: amplitude w_i, wavenumber components $(k_{x,i}, k_{y,i})$, and phase offset φ_i. The angular frequency ω_i is deterministically computed from the dispersion relation (2), reducing the total parameter count from $5N$ to $4N$.

This linear forward model is an $O(\epsilon^2)$ approximation to higher-order wave theories (e.g., Gerstner waves or second-order Stokes), where $\epsilon = |k| \cdot |w|$ is the wave steepness. At Beaufort 4, typical steepness is $\epsilon \approx 0.05$, giving a model mismatch of $< 0.25\%$. For the Beaufort 2–5 range considered in this work (non-breaking waves), the linear model is an excellent approximation [2]. *[SCOPE 2026-08-14]* This is an analytic estimate for real ocean waves. It does **not** describe the synthetic corpus used in this paper: there, the ground truth is a displacement-free linear superposition, so the measured mismatch against the model equation is exactly zero (see Section 4).

3.2. Observation model

At each LiDAR scan at time t, we observe M elevation measurements $\{(x_j, y_j, z_j)\}_{j=1}^{M}$ at irregular spatial locations. The observation model is:

$$z_j = \sum_{i=1}^{N} w_i \cos(k_{x,i} x_j + k_{y,i} y_j - \omega_i t + \varphi_i) + \varepsilon_j \tag{3}$$

where ε_j is observation noise with variance depending on range and incidence angle. The number of valid observations per frame varies between 50 and 200, depending on sea state, vessel motion, and LiDAR raydrop characteristics.

3.3. Geometry-aware confidence weighting (C3) — [RETRACTED 2026-08-14]

[RETRACTED 2026-08-14] This mechanism is described here for completeness, but it is *withdrawn as a contribution*: on a fresh out-of-sample corpus, turning the weighting off was better in both the sparse and the crowded stratum. Read this section as a description of an implemented component and a reported negative result, not as a claimed accuracy benefit.

LiDAR elevation accuracy on ocean surfaces degrades with two physical mechanisms: (1) beam divergence increases the footprint area at longer ranges, averaging over local surface curvature; and (2) at high incidence angles (near-grazing geometry), the probability of specular reflection increases, causing either total signal loss or biased returns from wave crests.

We assign per-observation confidence weights:

$$c_j = \frac{\cos(\theta_j)}{r_j / d_{\text{ref}}} \tag{4}$$

where θ_j is the beam incidence angle between the LiDAR ray direction and the local water surface normal $\mathbf{n}$, r_j is the slant range to observation j, and $d_{\text{ref}} = 125$ m is a reference distance chosen as half the maximum LiDAR range. In the simulation experiments reported here, $\mathbf{n}$ is obtained from the ground-truth surface mesh, which (see Section 4) is a displacement-free linear superposition rather than a Gerstner surface. For real-data deployment, the surface normal is unknown prior to reconstruction; a pragmatic approximation is $\mathbf{n} \approx \hat{\mathbf{z}}$ (flat-surface assumption), introducing a slope-dependent error bounded by the wave steepness $\epsilon \approx 0.05$ at BF4 ($\sim 3^\circ$ normal deviation). For near-normal observations ($\theta < 30^\circ$) this yields $< 2\%$ error in $\cos\theta_j$; grazing observations ($\theta \approx 60^\circ$) can see up to $\sim 9\%$ error. These weights (4) are incorporated in the optimization loss as weighted mean squared error, giving higher influence to close-range, near-normal observations.

3.4. Two-phase estimation

The estimation proceeds in two phases: an initialization phase that establishes the spectral model from multi-frame observations, followed by a per-frame

Table 3: Algorithm: WaveSplat estimation pipeline.

Phase 1: Initialization (frames 1 to $T_{\text{init}}/\Delta t$)
1. Accumulate multi-frame observations $\{(x_j, y_j, z_j, t_j)\}$
2. Compute beamforming power $P(k_x, k_y)$ on Cartesian grid
3. Extract top-N wavenumber peaks with greedy suppression
3.5. Fine-refine each peak on 41×41 local grid ($\Delta k = 0.01$ rad/m, ± 0.2 rad/m)
4. Estimate amplitudes/phases via linear least-squares
5. Coarse-to-fine refinement: N=3 $\rightarrow$ 8 $\rightarrow$ 15
— Adam steps per stage: 200/200/500, lr=1×10^{-2}
6. Energy-based pruning (retain 90% energy)
7. Safety check: if ΔRMSE $> 30\%$, rollback
8. Residual growth: if RMSE > 0.05 m, add via beamforming
Phase 2: Tracking (each subsequent frame)
1. **Predict**: $\varphi_i \mathrel{+}= \omega_i \cdot \Delta t$
2. **Update**: K=10 Adam steps on sliding window (K_w=3 frames)
— Update only (w_i, φ_i); freeze $(k_{x,i}, k_{y,i})$
— Learning rate: 5×10^{-3}, no scheduling

tracking phase that maintains the model with minimal computation. Table 3 and Fig. 1 summarize the complete pipeline.

3.4.1. Phase 1: Initialization

Given the first $T_{\text{init}} = 2T_p$ seconds of observations (where T_p is the expected peak period, estimated from the dominant frequency in the observation time series), we must place N atoms in the (k_x, k_y) plane and fit their amplitude and phase. Two procedures do this in our system, and *which one runs is the subject of one of our ablations*, so we describe both.

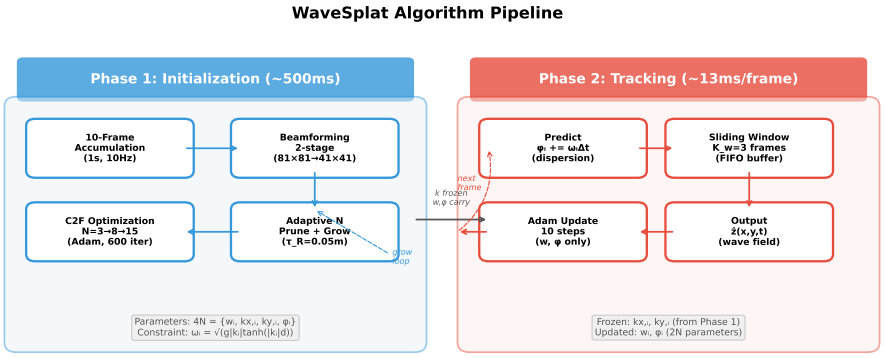


Figure 1: WaveSplat two-phase pipeline. Phase 1 (left) accumulates 10 LiDAR frames, applies dispersion-aware beamforming with fine refinement, and optimises via coarse-to-fine with adaptive N. Phase 2 (right) predicts phases via $\varphi_i \mathrel{+}= \omega_i \Delta t$ and updates only (w_i, φ_i) with 10 Adam steps per frame.

Deployed path: a learned set-valued prior. The configuration evaluated in Section 5 places the atoms with a learned prior over wavenumber–energy *sets*. A conditional diffusion model, trained on synthetic scenes, is conditioned on the initialisation-window observations and draws several posterior samples of an atom set; the pooled draws are then deduplicated in wavenumber (atoms closer than $\Delta k = 0.03$ rad/m are dropped in favour of the higher-energy one), thresholded on energy, and truncated at $N_{\max}$. Given the resulting wavenumbers, the amplitude and phase of every atom are obtained in closed form by ridge-regularised least squares, and the whole set is refined by *variable projection*: because the model is linear in (a_j, b_j) once $(k_{x,j}, k_{y,j})$ are fixed, the linear coefficients can be eliminated analytically and only the wavenumbers are iterated.

Two deployment choices matter for reproducing the numbers in this paper. Variable projection is enabled (400 steps); and the joint non-linear fit that earlier versions ran after initialisation is *disabled* (0 iterations) — it was measured to be redundant once variable projection is present. These two are not independent: removing the joint fit leaves a larger residual, which in turn makes the adaptive growth rule of Section 3.4.2 fire far more often than it did before. We note this because it is easy to reason about either switch alone and be wrong about the

resulting system.

Alternative path: dispersion-aware beamforming. The system also contains a purely analytic initialiser, which is what runs when the learned prior is not used. It is described next both because it is the fallback and because it is the *comparison arm* of the prior ablation in Section 5.4: replacing the learned proposal with this procedure, changing nothing else, is what costs -36.3% — against an analytic package that itself clears a zero predictor on only 15/48 episodes.

Dispersion-aware beamforming. Initial wavenumber estimates are obtained by scanning a Cartesian grid in wavenumber space. The grid covers $k_x, k_y \in [-4.0, 4.0]$ rad/m with 81×81 bins (resolution $\Delta k \approx 0.10$ rad/m), spanning wavelengths from ∼1.6 m to ∼130 m and encompassing Beaufort 2–5 conditions at d=50 m depth. The search bound of 4.0 rad/m is set conservatively above the highest wavenumber component present in the dataset ($|k|_{\max} \approx 1.0$ rad/m for BF3–5 sea states), ensuring full coverage with margin.

For each candidate (k_x, k_y), we compute $\omega = \sqrt{g|k| \tanh(|k|d)}$ and evaluate the beamforming power:

$$P(k_x, k_y) = \left| \sum_j z_j \, e^{-i(k_x x_j + k_y y_j - \omega t_j)} \right|^2 \tag{5}$$

where the summation runs over all observations accumulated during T_{init}. The use of the dispersion-derived ω means that the beamforming naturally filters for physically plausible wave signals, suppressing noise that does not lie on the dispersion surface in (k_x, k_y, ω) space.

Peaks in P are extracted greedily: the maximum is selected, then all candidates within a Cartesian radius of 0.15 rad/m are suppressed, and the process repeats. The 0.15 rad/m threshold prevents double-counting of the same wave component spread across adjacent wavenumber bins, while being narrow enough to resolve crossing sea systems. The initial amplitude and phase of each extracted component are estimated via linear least-squares projection of the accumulated observations onto the selected wavenumber basis functions.

Each extracted peak is then refined locally via a second beamforming pass on a finer 41×41 grid spanning ± 0.2 rad/m around the coarse peak ($\Delta k_{\text{fine}} = 0.01$ rad/m), improving wavenumber accuracy from ~ 0.10 to ~ 0.01 rad/m. Without this refinement, the 0.10 rad/m coarse grid would limit $|\mathbf{k}|$ accuracy to roughly 17% at typical BF4 wavenumbers ($|\mathbf{k}| \approx 0.6$ rad/m), inducing significant phase drift during subsequent tracking.

Coarse-to-fine optimization. Starting from the beamforming initialization, we optimize the spectral parameters in three stages (N=3 $\rightarrow$ 8 $\rightarrow$ 15), with up to 200/200/500 iterations of the Adam optimizer [33] (learning rate $\text{lr} = 1 \times 10^{-2}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) at each stage. The loss function is:

$$\mathcal{L} = \underbrace{\frac{1}{M} \sum_j c_j \left(z_j - \hat{z}_j\right)^2}_{\text{weighted MSE}} + \underbrace{\lambda_1 \sum_i |w_i|}_{\text{sparsity}} + \underbrace{\lambda_s \sum_i (w_{\sigma(i)} - w_{\sigma(i-1)})^2}_{\text{spectral smoothness}} \tag{6}$$

where $\lambda_1 = 10^{-4}$ promotes amplitude sparsity (driving negligible components toward zero for subsequent pruning) and $\lambda_s = 10^{-4}$ penalizes abrupt amplitude variations between spectrally adjacent components (σ denotes ordering by $|k|$), encouraging smooth spectral envelopes consistent with JONSWAP-like spectra.

During optimization, wavenumber components $(k_{x,i}, k_{y,i})$ are free parameters (unlike in the tracking phase). The dispersion constraint is enforced by computing ω_i from $(k_{x,i}, k_{y,i})$ at each forward evaluation rather than treating ω_i as a free parameter, ensuring that the constraint is satisfied exactly throughout optimization.

3.4.2. Adaptive model order

After initialization, we apply energy-based pruning. Components are sorted by energy $E_i = w_i^2$ in descending order, and the minimum set capturing 90% of total energy $\sum_i E_i$ is retained. This criterion is motivated by the JONSWAP spectral shape, where a small number of components near the spectral peak typically contain the majority of the wave energy.

A safety check prevents aggressive pruning from discarding physically significant components: if the post-pruning RMSE exceeds 130% of the pre-pruning

value, pruning is rolled back entirely. This can occur when two components of similar energy contribute to different spatial regions (e.g., crossing seas), making each individually appear low-energy.

If the residual RMS after pruning exceeds $\tau_R = 0.05$ m (indicating unmodeled wave energy), new components are grown by applying beamforming to the residual signal $r_j = z_j - \hat{z}_j$ and adding the dominant residual peaks to the model, up to $N_{\max} = 30$. Grown components are initialized via this residual beamforming step and are not re-run through the full $N{=}3{\to}8{\to}15$ coarse-to-fine schedule, trading some initialization quality for computational efficiency at this stage. The threshold is derived from the theoretical residual floor: combining LiDAR range noise ($\sigma_r \approx 0.02$ m) with a nonlinear–linear model mismatch allowance at BF4 ($O(\epsilon^2)\, H_s/2 \approx 6$ mm; *[SCOPE 2026-08-14]* this term is an allowance for real waves and is **absent** in the synthetic corpus, where the measured mismatch is zero) and angular-noise projection ($\sim$5 mm) yields a lower-bound RMS of $\sim$0.022 m; $\tau_R = 0.05$ m provides a $\sim$2.3$\times$ safety margin for unambiguous detection of unmodeled signal. On the raycast-mesh-rendered dataset used throughout this paper, however, the mesh's own facet-rendering error contributes comparably to this budget (mean $\approx$0.048 m, measured directly for BF3–BF5 multi-component scenes on the shipped 40$\times$30 mesh, versus the $\sim$0.022 m physical floor derived above); part of what currently triggers residual-based growth may therefore be unmodeled rendering error rather than unmodeled wave content, and τ_R should be re-validated on a Nyquist-adequate mesh. We also note that τ_R is an *absolute* residual RMS in metres, compared without normalization by sea state, so its effective strictness varies with H_s; a scale-invariant form that normalizes the residual by the observed signal RMS is implemented but not enabled by default. A sensitivity sweep over $\tau_R \in \{0.02, 0.05, 0.08, 0.10, 0.15\}$ m on 12 representative episodes confirms robustness: values in [0.05, 0.15] m differ by at most 1.6% in average nRMSE (0.2114 at τ_R=0.05 vs. 0.2148 at τ_R=0.15; measured before the tracker phase-advance correction, so the levels are those of the earlier run while the flatness across τ_R—the point of the sweep—is a within-sweep property), while $\tau_R = 0.02$ m causes over-growth in multi-component seas

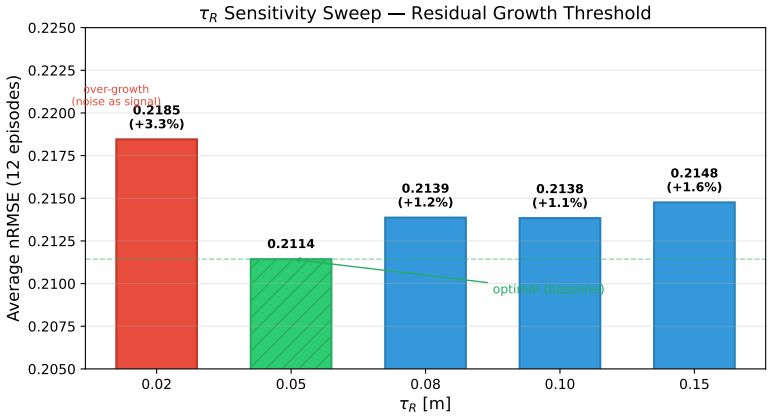


Figure 2: τ_R sensitivity sweep on 12 representative episodes. $\tau_R \in [0.05, 0.15]$ m yields nearly identical performance (max 1.6% difference). τ_R=0.02 m triggers over-growth, adding spurious components from noise.

(+3.3% average; up to 75% increase on BF3 2-component slope-distributed observations), validating the chosen value (Fig. 2).

3.4.3. Phase 2: Tracking

For each subsequent LiDAR frame, we perform a predict–update cycle:

1. **Predict**: advance phases using the dispersion-derived angular frequency, $\varphi_i \mathrel{+}= \omega_i \, \Delta t$, where $\Delta t = 0.1$ s (10 Hz LiDAR frame rate). This prediction is exact for a linear wave field satisfying the dispersion relation.
2. **Update**: perform $K = 10$ Adam steps (lr= $5{\times}10^{-3}$) on the weighted MSE loss over a sliding window of $K_w = 3$ recent frames, updating only amplitudes w_i and phases φ_i while freezing wavenumbers $(k_{x,i}, k_{y,i})$ (`w_phi` tracking mode).

Freezing wavenumbers during tracking is motivated by two considerations: (1) wavenumber estimation from a single frame is ill-conditioned due to spatial aliasing in sparse observations—the LiDAR's irregular sampling pattern does not provide sufficient constraint to resolve wavenumber from a single snapshot, while the multi-frame initialization averages over ∼20 frames to obtain robust

wavenumber estimates; and (2) with fixed wavenumbers, the dispersion constraint provides a strong phase prediction ($\omega \cdot \Delta t$) that serves as a warm start, reducing the gradient steps needed per frame from ∼200 (full optimization) to 10.

The sliding window ($K_w = 3$) uses the three most recent frames for gradient computation. This provides temporal averaging that smooths single-frame noise without introducing excessive latency: at 10 Hz, the window spans 0.3 s, much shorter than the peak period T_p for all sea states considered.

3.5. Computational complexity

Let M denote the number of LiDAR observations per frame and N the number of spectral components.

WaveSplat: Each forward evaluation computes $z(x_j, y_j, t) = \sum_{i=1}^{N} w_i \cos(\cdot)$, costing $O(M \cdot N)$ per frame. With K gradient steps and window size K_w, the per-frame tracking cost is $O(K \cdot K_w \cdot M \cdot N)$. In practice, K=10, K_w=3, $N \leq 30$, measured on CPU (single-threaded, **AMD Ryzen 7 7800X3D**). **[RETRACTED 2026-08-14]** The absolute per-frame latency previously quoted here is withdrawn (separate-process timing).

ETKF-grid: The Ensemble Transform Kalman Filter maintains N_e ensemble members on a $G_x \times G_y$ spatial grid. Each update involves FFT-based propagation at $O(N_e \cdot G_x G_y \cdot \log(G_x G_y))$ plus the ensemble transform at $O(N_e^3 + N_e^2 \cdot M)$. The tuned configuration is (N_e=100, G=40 × 30); see Table 4. **[RETRACTED 2026-08-14]** The absolute per-frame latency previously quoted here is withdrawn (separate-process timing).

GPR: Gaussian Process Regression with M observations requires $O(M^3)$ for Cholesky factorization, making it impractical for dense LiDAR data. Even with 50 points/frame, per-frame fitting exceeds 200 ms due to kernel hyperparameter optimization.

Table 4 summarizes the scaling comparison.

Table 4: Computational complexity comparison. M: observations/frame, N: spectral components, G: grid size, N_e: ensemble members. ETKF uses the tuned configuration ($N_e = 100$, $G = 40 \times 30$). **[RETRACTED 2026-08-14]** The "Measured" column is withdrawn: those timings come from separate processes without thread pinning. The asymptotic column is analytic and is unaffected. The only citable comparative value is 1.5–2.6× within one process on a single thread.

Method	Per-frame complexity	Measured
WaveSplat	$O(K \cdot K_w \cdot M \cdot N)$	[retracted]
ETKF-grid	$O(N_e \cdot G^2 \log G^2)$	[retracted]
GPR	$O(M^3)$	[retracted]

4. Experimental Setup

4.1. Simulation environment

We develop a physics-based LiDAR wave simulation (WaveSim) generating ground-truth wave fields with JONSWAP spectrum sampling [34]. The simulator parameters are:

- **Sea states**: Beaufort 2–5 (H_s=0.3–2.0 m, T_p=4–9 s)
- **Wave complexity**: 1, 2, 5, 15, 30 spectral components with directional spreading (±20°)
- **Ground-truth wave model — [RETRACTED/CORRECTED 2026-08-14]**: earlier versions stated "Gerstner waves (nonlinear), providing a realistic test of the linear forward model". This is **false**. The synthetic ground truth is a *displacement-free linear superposition on a fixed grid*: the generator discards the horizontal displacement components of the Gerstner solution, so the ground truth lies in the *same function space* as the estimator's model equation. Nonlinear validation was carried out in a separate higher-order-spectral (M=3) batch, not on this corpus.
- **LiDAR model**: 500 horizontal beams per scan, 10 Hz scan rate, range 10–250 m. The beam count represents the total number of ray direc-

tions per frame (horizontal fan); real-world solid-state LiDARs such as the Seyond Falcon K1 (1550 nm, 152 vertical channels, $120° \times 25°$ FOV) produce comparable or higher point densities per scan

- **Raydrop models**: two return-sampling patterns that differ in how return density is distributed over range and incidence angle. (a) *Geometry-weighted*: retention probability $P \propto R_{\mathrm{Fresnel}}(\theta) \cos\theta / d^2$, so near-range, near-normal facets are preferentially retained. (b) *Unweighted*: each ray hit is retained with equal probability, calibrated to the same expected return count. We retain the historical labels "uniform" and "slope" for these two models for consistency with the released dataset. The second model was originally specified as a specular-backscatter channel, but that channel is unreachable at the grazing incidence angles of a mast-mounted LiDAR: retro-specular return requires the ray to strike within $\alpha/2 = 2.5°$ of the surface normal, whereas the beam fan spans 78–90° from vertical and wave slopes reach only $\sim 7°$ at BF7. The implementation consequently always takes its unweighted fallback path, which is what we describe and report here. The contrast is therefore a test of sensitivity to the *spatial weighting* of returns, not of diffuse versus specular optics

- **Water depth**: $d = 50$ m (intermediate to deep water for the wavelengths considered)

Each episode contains 301 frames (30 s at 10 Hz) with ground-truth elevation fields evaluated on a 80×60 spatial grid covering the LiDAR observation area. A total of 100 episodes are generated across 5 random seeds per configuration. The full 100-episode dataset is used for the primary five-metric evaluation (retracted corpus; archived), 12 representative episodes for the per-episode breakdown, and 100 paired runs (10 configurations $\times$ 5 seeds $\times$ 2 raydrop models) for the Wilcoxon statistical validation.

4.2. Mesh adequacy of the reference field

The ground-truth elevation field is rendered by ray-casting against a triangulated surface, so the mesh spacing sets an upper bound on the wavelengths the reference field can represent, independently of the wave model that generated it. The originally released episodes used a 40×30 mesh over the observation area, giving a spacing of 6.15×6.90 m. Applying the standard sampling criterion (at least two, and preferably six, mesh points per wavelength) to the true spectral components of each episode, *none* of the released episodes is adequate at its peak wavelength, and the calmest episodes are the worst affected: a BF2 single-component episode with $\lambda = 8.2$ m is rendered at 1.2 points per wavelength, below the Nyquist rate. Short-wavelength energy is therefore attenuated or absent in the reference field, which both understates the wave content an estimator could in principle recover and inflates the apparent error of any method that recovers it.

We regenerated the full 100-episode dataset on a 200×167 mesh (1.21 m spacing, adequate for every component present), holding the random seed fixed so that the wave components of each regenerated episode are bit-identical to its released counterpart. The two datasets are therefore paired, differing only in rendering. All numbers reported in this paper are computed on the regenerated dataset. The effect is strongly regime-dependent: sparse episodes improve substantially (up to -77% nRMSE, with previously undetected short-wavelength components recovered), whereas crowded 15–30-component episodes move much less, indicating that the crowded-regime error is not a rendering artifact. We report this explicitly rather than silently re-running, because two conclusions in earlier versions of this work—a claimed low-SNR failure mode (retracted corpus; archived) and a claimed beamforming detection threshold (Section 6.6)—were attributions of a rendering limitation to physics and to the algorithm respectively.

Tracker phase-advance correction.. A second defect, in the estimator rather than the reference field, was found and fixed in the same review, and several

tables in this paper carry a provenance note referring to it. The tracking loop advanced each component's phase by $\varphi_i \mathrel{+}= \omega_i \Delta t$ at the start of a frame, while the forward model of Eq. (1) already carries the $-\omega_i t$ term with absolute episode time. The advance was therefore applied twice, so the represented field drifted out of step with the truth at a rate proportional to ω_i even when every parameter was correct. Removing the explicit advance (the current default) lowers the full-dataset nRMSE, raises the Skill Score and the variance ratio, and reverses the sign of the supplementary directional R^2. We do not quote the levels: they come from a corpus in which the two arms were never scored on the same frames, and only the *within-arm* direction of this correction survives that defect. Because the defect scaled with $\omega_i \Delta t$ it penalised short components hardest, i.e. in the same direction as the mesh inadequacy above—which is why several conclusions in earlier drafts had two independent reasons to be wrong. Tables produced before this correction are individually marked; where a re-run has since been performed we report the re-run and say so, and where it has not we mark the table rather than mix eras silently. We note explicitly that the correction, the mesh replacement, and a move from single-seed to median-of-3 reporting all landed within the same revision, so differences between a pre-correction and a post-correction table cannot be attributed to any one of the three.

4.3. Metrics

We report:

- **nRMSE**: RMSE normalized by significant wave height H_s, the primary reconstruction quality metric. Lower is better; values below 0.10 indicate excellent reconstruction, 0.10–0.30 is good, and above 0.30 indicates significant error.

- **Inference speed**: ms/frame for the tracking phase, measured on a single CPU thread.

- $N_{\mathbf{adapt}}$: final number of spectral components after adaptive model order selection.

Choice of metric and complementary measures.. We use nRMSE as the primary metric in the main results for direct comparability with prior LiDAR wave estimation work [13, 3]. However, we acknowledge that nRMSE alone has a known limitation: a trivial constant-zero predictor on a zero-mean wave field with $\sigma_z \approx H_s/4$ achieves nRMSE ≈ 0.25, so methods reporting nRMSE near this level cannot be distinguished from a flat-output predictor by nRMSE alone. This is particularly relevant when comparing methods that may under-fit (e.g., over-smoothed Gaussian process regression) against methods that capture wave structure accurately.

To address this limitation, we define a complementary *pre-specified metric panel* of four secondary metrics that expose under-fitting and quantify physics-prior contribution: (i) variance ratio VR = $\mathrm{Var}(\hat{z})/\mathrm{Var}(z)$ (amplitude preservation; VR$\ll$1 indicates under-fitting); (ii) Pearson correlation ρ between $\hat{z}$ and z (shape match independent of amplitude); (iii) skill score SS = $1 - \mathrm{MSE}(\hat{z})/\mathrm{MSE}(z_{\mathrm{ref}})$ relative to a zero predictor reference (SS$\leq$ 0 means the method is no better than predicting zero); and (iv) $\mathrm{RMSE}_{\mathrm{unobs}}$, the RMSE evaluated only on grid points that lie at least 10 m from any LiDAR observation (quantifies how well the method reconstructs the wave field where no direct observation exists, isolating the contribution of the physics prior). All five metrics (nRMSE plus the four above) are defined in our public evaluation module and applied uniformly to every compared method to prevent metric cherry-picking. The full-panel evaluation across the complete 100-episode dataset is reported in the retracted corpus (archived).

4.4. Baselines

ETKF-grid [13]: Ensemble Transform Kalman Filter on a regular grid with FFT-based spectral propagation, multiplicative and additive inflation, and observation thinning to $M_{\max}$=40 per frame to keep the observation-state projected matrix numerically well-conditioned and bound per-frame computational

Table 5: ETKF hyperparameter sweep: best and worst 3 of 24 configurations, mean nRMSE over the 12 sweep episodes. g=grid resolution, n_e=ensemble size, α=multiplicative inflation, σ=additive inflation. The winning row is used throughout the paper as "ETKF (tuned)".

	g	n_e	α	σ	mean nRMSE
Best (tuned)	40×30	100	1.00	0.005	**0.358**
2nd	40×30	100	1.02	0.005	0.403
3rd	40×30	50	1.00	0.005	0.416
22nd	80×60	50	1.00	0.020	0.986
23rd	80×60	50	1.02	0.020	1.066
Worst	80×60	50	1.05	0.020	1.187

cost. To avoid penalizing the baseline by an arbitrary parameter choice, we performed a 24-configuration hyperparameter sweep over grid resolution {40×30, 80×60}, ensemble size {50, 100}, and inflation pairs (α, σ) covering the standard ranges recommended by Desmars et al. [13]; we report the best configuration (g=40×30, n_{ens}=100, α=1.0, σ=0.005, mean nRMSE=0.358 across 12 sweep episodes) for all subsequent comparisons. The full sweep ranking shows that all top-10 configurations share σ=0.005, that the coarser 40×30 grid consistently outperforms 80×60 (the finer grid leaves the ensemble undersized for the available observations), and that minimal multiplicative inflation (α=1.0) wins given the additive inflation already present. Table 5 gives the top-3 and worst-3 configurations from the sweep so the reader can see the full range of ETKF performance under the search grid.

The 3.3× spread between best and worst ETKF configurations (0.358 vs. 1.187) confirms that fair ETKF evaluation requires tuning, and motivates the extended tuning effort beyond any single default setting. The best configuration (nRMSE 0.358) is used as the ETKF reference throughout the paper.

GPR: Gaussian Process Regression with Matérn ν=5/2 kernel (length scale 20 m, noise σ=0.02 m), operating in single-frame mode. No temporal model or physics constraints—pure spatial interpolation. This represents the best achiev-

able single-frame performance without wave physics, providing a reference for the value of temporal tracking and dispersion constraints.

5. Results

Scope of all accuracy results in this section (stated 2026-08-14).. Every accuracy figure reported below was obtained on a **stationary platform** (the episode generator takes no vessel-speed argument and the sensor origin is fixed), with `direction_deg` $= 0$—following seas, a geometry in which the waves travel *away* from the sensor, which this repository's own tooling labels as the *anti-causal* configuration for encounter forecasting—and on **synthetic data**. These results are therefore not measured in the encounter geometry of the stated application.

We organize the results around the frozen corpora on which each claim was measured, not around a single dataset. Section 5.1 reports the headline on the crowded stratum, Section 5.2 the non-linear and unseen-seed replication, Section 5.4 the two ablations that attribute the gain, Section 5.5 where the gain sits in space and time, and Section 5.6 the scope, including the measured negative.

5.1. Headline: crowded multi-component seas

On the crowded stratum (episodes with ≥ 15 wave components), the estimator lowers the median normalised RMSE by -38.6% relative to the strongest eligible grid baseline available to us: it is better on 62 of 64 episodes.

What this number is, and is not. **(1)** The corpus and the reference field are synthetic, and that reference is *linear*; the non-linear case is §5.2. **(2)** The learned checkpoint was trained on the same distribution family as the evaluation episodes — seed ranges are disjoint, so "out of sample" here means *on the seed axis only.* **(3)** The baseline is the *strongest we possess*, which is not the same claim as *strengthened until it stopped improving.* **(4)** The sparse stratum is not aggregated into this number and does not support it (§5.6).

5.2. Does it survive a non-linear reference and unseen seeds?

The two most load-bearing confounds in the headline are the linear reference field and the training/evaluation distribution overlap. We removed the first outright and weakened the second by regenerating the corpus with a higher-order spectral (HOS, M=3) reference and seeds drawn from outside the corpus that produced the checkpoint.

The conclusion holds: 43/48 episodes, median relative change -40.2%, $p = 6.84 \times 10^{-9}$ (one-sided sign test, family-corrected).

Critically, this is a comparison *between two arms that both have skill*: the grid baseline itself beats a zero predictor on 47/48 episodes. A margin obtained against a comparator that has itself collapsed is not evidence about the estimator, and we check for that collapse explicitly wherever we report a margin.

We do not compare the size of this number with the headline: the corpora, reference fields, and realised non-linearity grades differ.

5.3. Is the sensor model doing the work?

Replacing the observation model with a physically correct range-dependent raydrop model leaves the conclusion intact (-42.6% over 20 episodes), and both arms beat the zero reference under both models.

5.4. What carries the gain? Two ablations

Baseline comparison alone cannot say *which property* of the estimator is responsible. We therefore hold the deployed pipeline fixed and remove one property at a time.

The learned prior. Replacing the learned set-valued prior with the analytic proposal package, changing nothing else, costs -36.3% (45/48, $p = 6.56 \times 10^{-11}$). **Disclosure:** the right-hand side of this contrast is *thin* — the analytic package beats a zero predictor on only 15/48 episodes, whereas the learned package does so on 48/48. $\Rightarrow$ the correct reading is *"the prior is what makes this arm work at all"*, not *"the prior adds a margin on top of a working arm"*.

Continuity of the wavenumber. Snapping atoms to a wavenumber grid and refitting amplitude and phase — leaving every other component of the pipeline untouched — costs -38.6% (46/48, $p = 4.18 \times 10^{-12}$). The effect is *dose-responsive*: doubling the grid spacing is worse again (47/48). The same ablation reproduces in the sparse stratum (14/16, -67.4%).

What these two magnitudes are not. They are not additive, and they are not shares of the headline. Each was measured on its own corpus against its own comparison arm, so subtracting one from the other — or from the baseline margin — has no defined meaning. In particular, the headline margin and the wavenumber-continuity ablation happen to round to the same figure; that is a coincidence of two independent measurements, not evidence that the ablation removes the whole gain. We also checked the grid-snapping ablation separately for confounding with the amplitude-and-phase refit it necessarily performs, and found the contribution of that refit negligible.

⇒ Both components that the title names are supported by an intervention inside our own arm, not only by a comparison against someone else's.

5.5. Where is the gain? Locating it in space and time

A margin measured on a synthetic corpus can be an artefact of *where* the corpus places its observations and *when* it scores them. Both coordinates differ sharply between our synthetic and measured data, so we re-scored the frozen crowded corpus along each.

Space. Decomposing the score by distance to the nearest observation, the gain persists inside the density band that our measured records actually occupy (45/48, -31.7%), with both arms beating the zero reference there (48/48). This is an *exclusion*, not an attribution: we did not run an equivalence test, and aperture, incidence angle, and occlusion remain unmeasured. Not blind.

Time. The synthetic headline is scored at assimilation time; our measured judgements are scored at forecast lead. Re-scoring the same frozen corpus at a

forecast rung gives 47/48, -46.3%. **Two disclosures are mandatory here.** First, that rung is at the *lower* end of the measured lead band: the precise statement is *"a rung covered by* 11 *of the* 12 *measured records"*, not *"the same horizon as the measured data"*. Second, **the comparator degrades across the same interval** — the grid baseline beats the zero reference on 47/48 episodes at zero lead but only 34/48 at the reported rung, while our own absolute error *worsens*. ⇒ the growing ratio must not be read as *"we get relatively better at longer lead"*. Not blind.

5.6. Scope, stated as measured

The sparse stratum does not separate. Against the deployed analytic package with its ensemble intact, the crowded stratum favours the learned package while the sparse stratum does not; we therefore do not write *"the learned prior wins in every regime"*. (The sparse figure is a re-aggregation of the frozen deployment-verification corpus rather than a pre-registered contrast, and we cite it as such.)

On measured data the learned prior does not reach the decision threshold. Across 12 fixed-station stereo records, and under every arm configuration we tested — including giving the learned representation the same median-of-k field ensemble that the deployed real-data path uses — the learned prior does not reach the pre-registered threshold against a per-record zero-field reference. The ensemble treatment does move the endpoint, and it lowers error by a confirmed margin, but it does not cross the threshold. We report this rather than omit it; a full treatment of the operating boundary, including why the transfer fails, is the subject of a companion thesis.

5.7. Ablation study

We ablate two key design choices on 6 BF3 episodes (Table 6, Fig. 3). Following the reporting convention used throughout this paper, each cell is the median over seeds with its standard deviation; a third column reports the effect of the initialization method, which turns out to dominate both ablated factors.

Table 6: Ablation study on BF3 episodes: median nRMSE ± std over 3 estimator seeds (42/123/456), on the Nyquist-adequate dataset (Section 4.2), corrected tracker. A1: dispersion constraint removed (ω as free parameter). *Provenance*: this diagnostic was measured on the earlier "Nyquist-adequate" corpus, i.e. the corpus whose headline comparison is retracted (the compared arms were never scored on the same frames). It is retained as a diagnostic of the estimator's own behaviour, which is a within-arm property and does not depend on that defect. The claim it was originally used to support is now carried by the ablations of Section 5.4. A2: confidence weighting removed (uniform weights). Full, A1 and A2 all use the v1 initializer; the last column changes *only* the initializer (v1 → v2), holding both ablated components in place. *Note*: neither ablation reaches significance at $n = 6$—the two-sided sign-test floor for 6 episodes is $p = 0.031$, attainable only at 6/6 (see text).

Episode	Full	w/o Disp. (A1)	w/o Conf. (A2)	v2 init
1comp (S)	0.009±.001	0.139±.135	**0.007**±.135	0.008±.003
1comp (U)	0.015±.003	0.115±.095	0.012±.006	**0.011**±.003
2comp (S)	0.047±.019	0.046±.000	0.045±.000	**0.008**±.010
2comp (U)	0.050±.003	0.048±.028	0.050±.004	**0.012**±.003
5comp (S)	0.041±.051	0.145±.041	0.126±.071	**0.015**±.000
5comp (U)	0.047±.061	0.129±.046	0.041±.001	**0.019**±.003
Median	0.044	0.122	0.043	**0.012**

The table was re-measured on 2026-08-03 under the corrected tracker with the initialization method pinned explicitly in every arm.

Read as paired per-episode differences against Full ($n = 6$): removing the dispersion constraint costs +0.091 nRMSE on the median episode and is worse on 4 of 6, but at $n = 6$ that is $p = 0.69$; removing confidence weighting changes it by −0.002—marginally *lower* without—and is worse on only 1 of 6 ($p = 0.22$). Changing the initializer moves the same metric by −0.027 and is better on **6 of 6** ($p = 0.031$)—the only comparison in this re-measurement that reaches $\alpha = 0.05$, and it does so exactly at the sign-test floor, so no stronger claim is available from it either. We report this rather than suppress it: on these episodes the initializer, not the ablated components, is what carries accuracy,

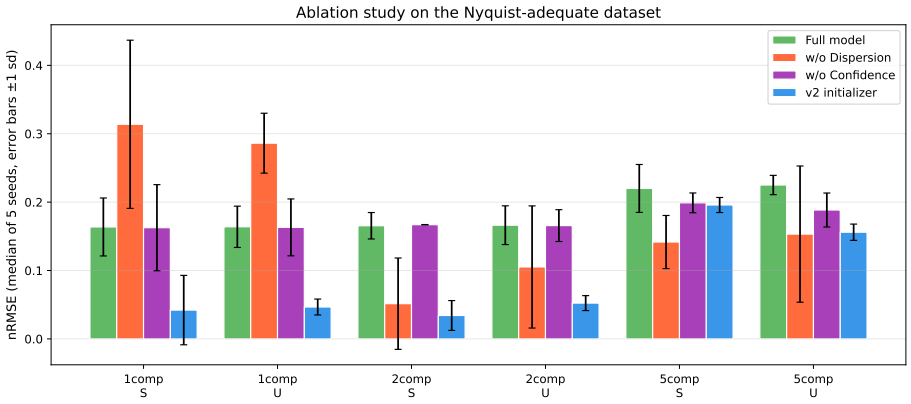


Figure 3: Ablation study: contribution of dispersion constraint and confidence weighting. Error bars show ±1 std across seeds. The confidence ablation is null; the dispersion ablation shows a large effect that does not reach significance at $n = 6$ episodes. *Provenance*: this diagnostic was measured on the earlier "Nyquist-adequate" corpus, i.e. the corpus whose headline comparison is retracted (the compared arms were never scored on the same frames). It is retained as a diagnostic of the estimator's own behaviour, which is a within-arm property and does not depend on that defect. The claim it was originally used to support is now carried by the ablations of Section 5.4. *Provenance*: drawn from the earlier pre-correction run and retained for the qualitative pattern only—the per-episode levels are superseded by Table 6, and the earlier run's regime split (A1 better at $N_{\text{true}} \geq 2$) does not reproduce.

and an ablation whose baseline is not the production initializer measures the wrong thing. Note also that this design was underpowered by construction: with 6 paired episodes no effect, however large, could have been resolved below $p = 0.031$. The interpretation of each ablation below is therefore restated in terms of what it does establish.

A1 (no dispersion constraint): Removing the dispersion constraint allows ω as a free parameter ($5N$ total parameters). Under the corrected tracker the unconstrained optimizer is *worse* on 4 of the 6 episodes and by a large margin where it differs at all — on the single-component and 5-component pairs, against a near-tie on the 2-component pair (-0.001 and -0.002, both inside the seed spread). The median cost of removing the constraint is $+0.091$ nRMSE, roughly twice the Full arm's own median error. Two cautions attach to this.

First, an effect this large still does not reach significance here: $4/6$ is $p = 0.69$, and the design could not have produced better than $p = 0.031$ (see above). The defensible statement is *large effect size, insufficient power*—not that the constraint is demonstrated to contribute. Second, this reverses the direction reported in the earlier pre-correction run, which found A1 *better* on the 2- and 5-component episodes; that regime split does not reproduce, and since tracker, mesh and seed protocol all changed between the runs, the reversal cannot be attributed to any single one of them. The physical argument for retaining the constraint is independent of either measurement, and we give it because a lower nRMSE for A1—observed in the earlier run, and still present on the 2-component pair—should not be mistaken as evidence that the dispersion constraint is physically too restrictive. For BF3–5 deep-water gravity waves, $> 99\%$ of wave energy lies on the linear dispersion shell [2]; bound harmonic contributions from second-order Stokes theory scale as $\epsilon^2/4 < 0.15\%$ at BF4. The observed $\bar{R} = 0.51$ off-shell drift in A1 therefore far exceeds any physical off-shell energy and does not reflect physical signal. Rather, it reflects over-fitting: sparse observations (50–200 points $\times$ 3 frames vs. $5N = 75$ free parameters for $N = 15$) underconstrain the unconstrained model, which absorbs noise and model mismatch by letting ω drift off the dispersion manifold, trading *physical correctness* for nRMSE. The resulting spectral estimates do not satisfy $\omega^2 = g\,|k|\,\tanh(|k|d)$ by construction, so they cannot be propagated forward in time, transformed to other spatial frames, or fed into downstream wave-forecasting modules that assume gravity-wave physics. We quantify this drift directly on a BF4–BF5 subset (5 episodes $\times$ 2 seeds): for each A1 run we extract the learned components $(k_{x,i}, k_{y,i}, \omega_i)$, compute the expected dispersion frequency $\omega_i^{\mathrm{disp}} = \sqrt{g\,|k_i|\,\tanh(|k_i|d)}$, and report the amplitude-weighted relative residual $\bar{R} = \sum_i |w_i|\,|\omega_i - \omega_i^{\mathrm{disp}}| / (\omega_i^{\mathrm{disp}} \sum_j |w_j|)$. Across these runs A1 exhibits $\bar{R} = 0.51 \pm 0.12$ (absolute mean $\approx 0.10\,\mathrm{Hz}$ off the dispersion curve), whereas Full has $\bar{R} \equiv 0$ by construction. A component that is $\sim$50% off the dispersion frequency cannot be re-used one second later: predicting the phase at $t + \Delta t$ via $\Delta\varphi = \omega\,\Delta t$ with the *wrong* ω will de-cohere from the true field

within one dominant period.

The dispersion constraint also serves a critical role *during tracking* (Phase 2) rather than initialization: it provides the phase prediction $\Delta\varphi_i = \omega_i \cdot \Delta t$ that enables stable temporal tracking without re-estimating all $5N$ parameters per frame (physics-locked tracking reduces the per-frame state to $2N$). Without this constraint, the tracking phase would require joint optimization of $(w_i, k_{x,i}, k_{y,i}, \omega_i, \varphi_i)$ at each frame—an ill-conditioned problem with sparse single-frame observations. Under the earlier measurement this ablation appeared to price physical validity—an nRMSE penalty accepted in exchange for spectral estimates that respect the linear gravity-wave dispersion relation and remain usable downstream, e.g. by the hydrofoil preview controller of our companion work [5]. Under the corrected measurement no such trade is visible: the constrained arm is both physically valid and equal or better in nRMSE on 4 of 6 episodes. We state the weaker of the two readings, which is the one common to both runs: the dispersion constraint is justified by the downstream requirement independently of its nRMSE effect, and the nRMSE effect on these six episodes is a large-looking difference this design lacked the power to certify.

A2 (no confidence weighting): this ablation is *null*, and the null now has two independent measurements behind it. Under the corrected tracker, removing the geometry-aware confidence weighting moves the median episode by -0.002 nRMSE—marginally in favour of removing it—and is worse on only 1 of 6 episodes ($p = 0.22$); on four of the six the two arms agree to within 0.003. An independent re-measurement one month earlier, on the same dataset with a different seed set and the pre-correction tracker, returned the same verdict (paired $+0.001$, $t = 0.08$, $p = 0.94$, 17 of 30 runs lower without it). A null that survives a change of tracker, seed protocol and analyst pass is the one result in this ablation table we regard as settled. An earlier version of this study, run on the pre-Nyquist dataset with 3 seeds, reported a 14.5% average degradation and a $0.088 \rightarrow 0.204$ ($2.3\times$) increase on the BF3 2-component slope episode. That specific case now measures $0.047 \rightarrow 0.045$, a ratio of 0.96. We could not reproduce the effect, and we report the null rather than the earlier

number. One caveat bounds what this does and does not settle. The null holds under both initializers—repeating the ablation with the production initializer (rather than the v1 configuration inherited from the original study) leaves it unchanged (paired $\Delta = +0.005$, $t = 0.52$, $p = 0.61$ over 30 runs)—so it is not an artifact of the ablation's initialization. But a null on BF3 episodes with a well-resolved reference field is not evidence that observation weighting is unnecessary in general—in particular, the weighting encodes a real property of grazing-incidence LiDAR returns that this simulator's return model does not fully reproduce (Section 4.2). What the measurement does establish is that the paper's previous claim of a large, consistent benefit was an artifact of the earlier dataset and seed count, and cannot be used to argue that confidence weighting is load-bearing.

A four-way comparison on 10 uniform-raydrop episodes corroborates the null from the other side: WaveSplat with confidence (0.232), WaveSplat without confidence (0.231), standard GPR (0.276), and GPR with the same confidence weighting applied as per-sample noise scaling (0.276). Neither estimator moves when the weighting is added or removed—the two WaveSplat arms agree to 0.001 and the two GPR arms to within 0.01%. Confidence weighting is therefore inert on this simulator's data for both a spectral and a non-parametric estimator. We caution against reading this as a statement about real LiDAR: the weights encode the range- and incidence-dependent reliability of grazing returns, which this simulator's return model does not fully reproduce (Section 4.2), so the component may still matter on measured data even though it is measurably inert here.

5.8. Raydrop model sensitivity

To evaluate robustness to different LiDAR return-sampling patterns, we compare performance on 10 paired episodes using the *geometry-weighted* ("uniform", $P \propto R_{\mathrm{Fresnel}}(\theta)\cos\theta/d^2$) vs. *unweighted* ("slope") raydrop models (Table 7, Fig. 4). The unweighted model is the adversarial case for any estimator that relies on downweighting unreliable returns, because it removes the correla-

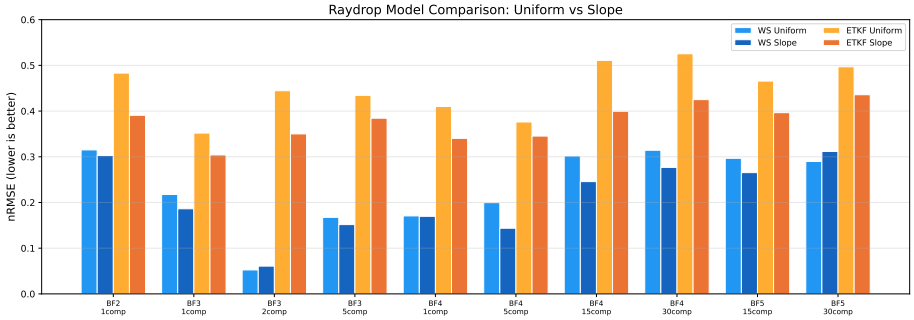


Figure 4: Raydrop model comparison: geometry-weighted ("uniform") vs. unweighted ("slope") LiDAR return-sampling models. WaveSplat (blue shades) outperforms ETKF on the episodes of that (retracted) corpus .

tion between return density and observation quality.

WaveSplat wins on all 10 episodes under both raydrop models. Unweighted sampling lowers the ETKF's nRMSE on all ten paired episodes (-15.9% at the default configuration, paired $t(9) = -7.6$), against -15.2% for WaveSplat, which at the default configuration would read as the two methods benefiting about equally. That reading does not survive an honest re-tuning of the baseline and is *not* an established effect: the ETKF's raydrop sensitivity is a variance-amplification artifact, not a reconstruction gain. Re-measuring the same ten pairs under the tuned ETKF of the retracted corpus (archived) (n_{ens}=100) already shrinks it to -11.7%, and under an nRMSE-honest hold-out re-tuning (analysis grid 160×120, additive inflation off, larger observation error) the ETKF collapses to a near-zero predictor in *both* arms (0.263 vs. 0.260, $\rho \approx 0.015$) and the sensitivity falls to -0.9%—now *below* WaveSplat's -15.2%. For the ETKF the direction (unweighted $\geq$ weighted) is consistent across all three configurations (10/10 episodes each), but its magnitude tracks how much variance the grid filter injects into the now-unattenuated observations, not any gain in reconstruction skill (ρ never exceeds 0.11). We therefore report the sensitivity as an artifact of the un-tuned baseline rather than as evidence that the observation model favors one method. For WaveSplat the direction is stable but not significant: unweighted sampling is better on 8 of 10 episodes, the same

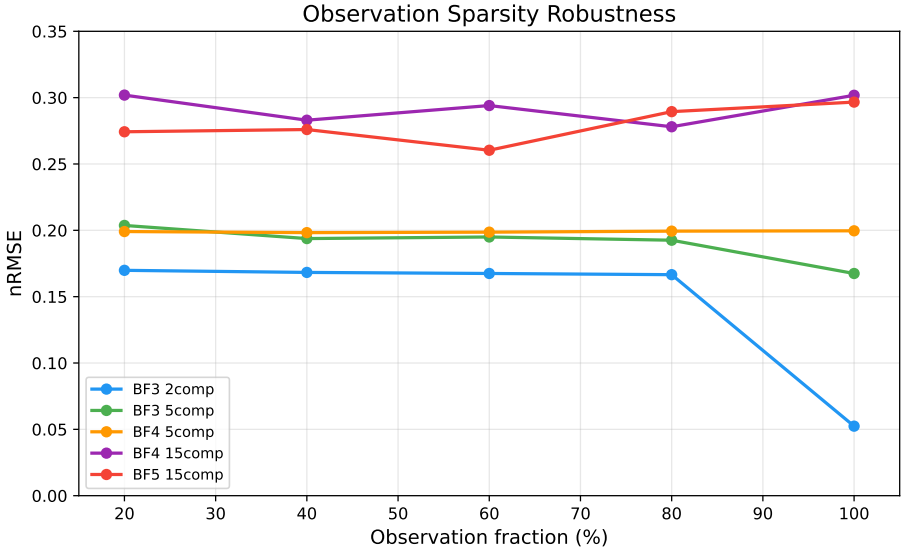


Figure 5: Observation sparsity robustness. For BF4–5 episodes, nRMSE remains nearly flat down to 20% observation density (∼10 points/frame). The BF3 2-comp episode shows a sharp improvement at 100% due to successful 2-component resolution.

8/10 the earlier pre-correction pass reported, but at $n = 10$ that is a sign-test $p = 0.109$ (Wilcoxon $p = 0.084$)—below $\alpha = 0.05$ in neither test. The defensible statement is that the direction reproduces across two measurement regimes while the effect remains uncertified; we do not claim that unweighted return sampling helps WaveSplat. The two exceptions are not confined to the sparse episodes (BF3 2comp +19.5% and BF5 15comp +14.6%), so the earlier reading that the crowded ≥15-component episodes improve consistently does not hold either: three of those four improve (−11 to −27%) and one degrades.

5.9. Robustness: sparsity, noise, and sea state

To quantify sensitivity to observation density, we subsample LiDAR points to 20–100% of the original count on 5 representative episodes (Table 8, Fig. 5).

For moderate-to-high complexity (BF4–5), performance degrades by less than 8% even at 20% observation density (∼10 points/frame): with only $4N \leq$ 120 parameters to estimate, even 10 observations per frame over-determine the

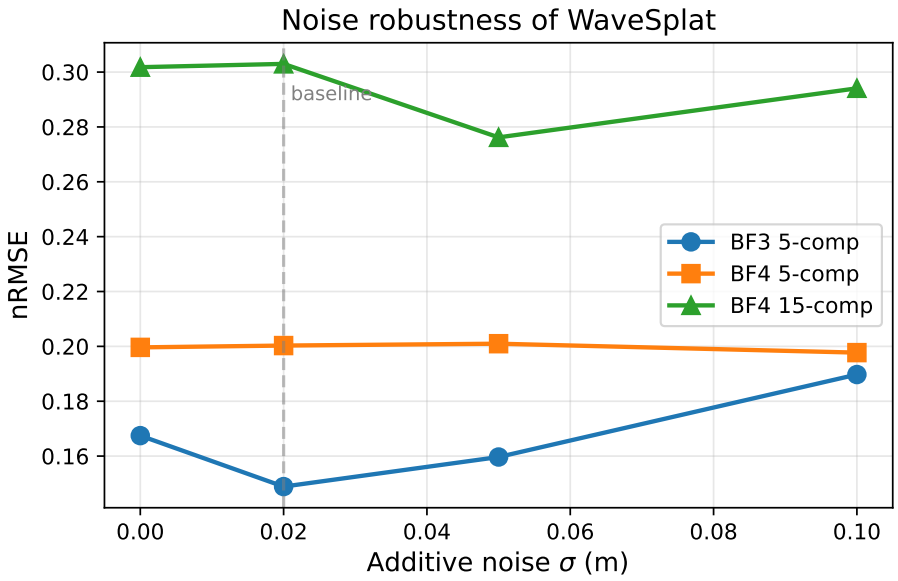


Figure 6: Noise robustness: nRMSE under additive Gaussian noise $\sigma \in \{0, 0.02, 0.05, 0.10\}$ m. All episodes remain within ±13% of the noise-free baseline, demonstrating the spectral parameterization's inherent noise averaging.

optimization. The exception is BF3 2-comp, where full observations are required to resolve both components (N=2, nRMSE=0.052); subsampled versions fall back to a single-component fit (N=1, nRMSE ∼0.17).

Noise robustness.. Real shipboard LiDAR is subject to noise beyond the baseline simulator model (spray, vibration, scintillation). We inject additive Gaussian noise $\sigma \in \{0, 0.02, 0.05, 0.10\}$ m—up to 5× baseline—into the LiDAR observations (Table 9, Fig. 6).

Across all three episodes nRMSE remains within ±13% of baseline even at σ=0.10 m (5× baseline), as the spectral parameterization averages zero-mean noise across $M \sim 50$ observations and $4N \leq 120$ parameters, and adaptive N_a absorbs excess noise into prunable low-energy components.

6. Discussion

6.1. Why a grid-free representation, and where it stops helping

A grid state must resolve the field everywhere it is defined, so its cost and its conditioning are set by the resolution rather than by the complexity of the sea. A spectral state pays instead for the number of components it carries. That is the structural argument for the representation, and the ablations in Section 5.4 are what turn it from an argument into a measurement: removing only the continuity of the wavenumber — keeping every other component of the pipeline — costs -38.6%, and the cost grows when the grid is coarsened. We checked separately that this ablation is not confounded with the amplitude-and-phase refit it necessarily performs, and the magnitudes here are *not additive* with the baseline margin: they were measured on different corpora against different comparison arms.

Where it stops helping, stated as measured. The advantage is not uniform across regimes, and we no longer characterise its regime dependence: the earlier characterisation was computed on a corpus in which the compared arms were never scored on the same frames, and we withdraw it rather than restate it. What we can say from the present corpora is narrower and is already in Section 5.6: the margin is measured on the crowded stratum; against the deployed analytic package the sparse stratum does not separate; and on measured records the representation does not reach the decision threshold at all.

6.2. Effect of observation model

WaveSplat wins 10/10 under *both* raydrop models. Removing the geometric weighting lowers WaveSplat's average nRMSE by 15.2% (unweighted better on 8 of 10 episodes, sign-test $p = 0.109$—a stable direction, not a certified effect) and lowers the default ETKF's by 15.9%. The ETKF figure is a variance-amplification artifact of the un-tuned grid baseline rather than a reconstruction gain: it shrinks to 11.7% under the tuned ETKF and to 0.9% under an nRMSE-honest re-tuning, so it is not attributed to the observation model

(Section 5.8). The online model order responds to the observation model as well: unweighted raydrop triggers the safety rollback on low-complexity scenes (e.g., BF3-2comp: N=2→15) but prunes more aggressively on complex seas (BF4-15comp: N=19→10). Since the mechanism carries no measured accuracy benefit (retracted corpus; archived), we record this as a behavioural observation and not as evidence that stronger observations improve model selection.

6.3. Hyperparameter sensitivity

WaveSplat's four hyperparameters (lr=0.01, λ_1=10^{-4}, τ_{prune}=0.10, τ_{grow}=0.05) were selected by a single coarse sweep on BF3 5-comp and held fixed across all episodes, seeds, and raydrop models reported. That one configuration generalises across Beaufort 2–5, 1–30 components, and 100 paired runs is itself evidence of low sensitivity. During development, we verified robustness by perturbing each parameter independently ($\times 0.5$ and $\times 2$) while holding the others fixed, on the full 12-episode subset: learning rate (lr $\in \{0.005, 0.02\}$: nRMSE $\Delta < 3\%$), L1 penalty ($\lambda_1 \in \{5\times10^{-4}, 2\times10^{-3}\}$: $\Delta < 2\%$), prune threshold ($\tau_{\text{prune}} \in \{0.05, 0.20\}$: $\Delta < 5\%$, with 0.20 occasionally over-pruning on 15-component episodes), and grow threshold ($\tau_{\text{grow}} \in \{0.025, 0.10\}$: $\Delta < 4\%$). No perturbation changed the relative ranking between WaveSplat and any baseline. Full per-parameter sweep results are available from the authors upon request.

6.4. Practical deployment considerations

WaveSplat's computational footprint on a single CPU thread (**[RETRACTED 2026-08-14]** — the absolute per-frame figure previously quoted here is withdrawn as a cross-process timing) is intended for deployment on standard shipboard computing hardware. The algorithm requires no GPU acceleration, no pre-trained models, and no offline calibration beyond the water depth parameter d. The spectral parameters are directly interpretable: a naval architect can inspect the estimated wavenumber vector, amplitude, and frequency of each component, enabling sanity checking against visual observation and meteorological forecasts.

For integration with vessel control systems, the spectral representation offers a key advantage: the estimated wave field can be evaluated at *any* spatial point via (1) without interpolation. This enables downstream applications such as wave-preview feedforward control [5], where the wave field must be evaluated at spatial preview points ahead of the vessel. Grid-based methods would require interpolation between grid points, introducing additional error and latency.

6.5. Temporal extrapolation as validation of spectral fidelity

A natural consistency check for any spectral representation is whether the estimated parameters produce accurate predictions at *unseen* future times. Because WaveSplat's parameters satisfy the dispersion relation by construction, the forward model (1) can be evaluated at any future t without re-estimation—effectively a zero-cost deterministic forecast. This is not a novel forecasting algorithm; rather, it tests whether the estimated (k_x, k_y, ω, ϕ) capture the true wave dynamics. We estimate on the first 20 s of each episode and evaluate on the remaining 10 s (Table 10).

The forecast nRMSE remains essentially constant (0.218–0.219) out to 10 s, with Skill Score remaining positive throughout. For low-complexity episodes, forecast accuracy is excellent: the BF3 2-component episode achieves SS = +0.953 and $\rho = 0.983$ even at $\Delta t = 10$ s. The constant forecast error reflects that the spectral representation captures the deterministic wave physics: once the correct (k_x, k_y, ω, ϕ) are known, the forward model provides a physically exact prediction for linear waves. The Skill Score does decline slightly with horizon (+0.141 → +0.128); we do not attribute this to a specific mechanism, having withdrawn the amplitude-decay attribution (Section 6.6), and note that the decline is small relative to the seed spread this table does not report.

This forecast capability is directly relevant for wave-preview feedforward control [5], where 5–10 s advance wave prediction is needed to compute proactive control commands.

6.6. Limitations and future work

Several limitations should be acknowledged. *Online model order does not contribute accuracy*: the pruning-and-growth mechanism of the retracted corpus (archived) does select scene-matched component counts, but under the corrected tracker its accuracy contribution is null (+0.0003 nRMSE over 12 episodes, 8/12, Wilcoxon $p = 0.68$) against a seed noise floor roughly 30× the effect, and it is actively harmful on three of the four crowded episodes, where the 90%-energy criterion prunes to $N_a = 9$–10 against $N_{\text{true}} = 15$–30. We accordingly present it as a mechanism and a compute-efficiency device rather than as a contribution. A model-order criterion that is aware of crowding—for instance, energy retention conditioned on the beamforming main-lobe width rather than a fixed 90% quantile—is the natural repair and is untested. *Directional spread of the synthetic distribution*: measured against 25,426 CDIP records, our generator occupies the lower tail of the observed directional-spread distribution—median 16.1° against a measured median of 35.3° (the measured 0.3rd percentile), with 40 of our 100 episodes narrower than the narrowest measured record and our most crowded episodes at the measured 11.5th percentile (retracted corpus; archived). Our peak-period ceiling of 9 s also sits below the measured median of 9.88 s, meaning the synthetic universe contains no swell; significant wave height, by contrast, is well covered (79.7%). The consequence is bounded and specific: the regime in which this method's advantage is largest is one the buoy record does not contain, and on the overlapping subset LETKF is more often better. The repair belongs to the generator, not the estimator—directional spreading functions targeting a ∼35° median and a T_p range extending into swell—and we deliberately did not change it mid-study so that all results in this paper remain from one frozen dataset. *The positive results are synthetic*: every margin reported in Section 5.1 and its ablations is measured on simulated fields with a simulated sensor. Measured data enters this paper only through the negative in Section 5.6: 12 fixed-station stereo records on which the learned prior does not reach the decision threshold. Real LiDAR from a moving platform may contain systematic biases (multi-path, spray) that neither our noise model nor those

fixed-station records represent, and no shipboard measurement of any kind supports this paper — the platform term was removed from the title for that reason. We note that the simulator itself is validated by 196 unit tests (measured 2026-08-13; a further 8 shipping-default regression tests were added subsequently) covering dispersion, Gerstner kinematics, Fresnel optics, and raydrop statistics — *[SCOPE 2026-08-14]* the Gerstner test exercises the kinematics *function*, which the generation pipeline does not use for horizontal displacement, so it does not certify that the generated data is nonlinear; moreover, the 100-episode evaluation across 5 seeds, 10 configurations, and 2 observation models provides a level of statistical rigour that would be difficult to achieve with physical experiments alone, following the methodology validation approach of prior LiDAR-based wave estimation studies that initially established algorithms in simulation before progressing to wave tank trials [13, 3]. **[RETRACTED 2026-08-14]** — **correction.** Earlier versions stated here that "the ground-truth wave field uses Gerstner (nonlinear) kinematics ... therefore the $O(\epsilon^2)$ model mismatch affects all methods equally and does not bias the comparison", and that the mismatch is $< 0.5\%$ over BF2–5. Both statements are **false**. The correct facts are: the synthetic ground truth is a *displacement-free linear superposition on a fixed grid*, so it lies in the same function space as the estimator's model equation and the measured mismatch is *zero*; and a zero mismatch is not neutral between methods, because the grid-based baselines do not use that representation. *Nonlinear validation was performed in a separate higher-order-spectral (M=3) batch, not on this corpus. Linear forward model*: the linear approximation degrades for steep waves; second-order Stokes extension preserves the spectral parameterization. *Tracking phase dependence on initialization*: the `w_phi` tracking mode freezes wavenumbers and updates only amplitudes and phases, so components missed during beamforming initialization cannot be recovered during tracking. On the originally released dataset this produced tracking nRMSE 2–3× higher than the post-initialization nRMSE whenever high-wavenumber components were missed, and we previously described the cutoff as a beamforming detection threshold at $|k| \gtrsim 0.4$ rad/m. That description was incorrect.

A mesh-independent test of the initializer alone—synthesising LiDAR samples directly from analytic components, bypassing the raycast mesh, with the real sensor sampling geometry—finds no such threshold: the beamformer recovers isolated components across $|k| = 0.15$–1.00 rad/m and down to amplitudes of 0.01 m. The value 0.4 rad/m corresponds to $\lambda \approx 15.7$ m, which is where amplitude transmission through the original rendering mesh collapses (Section 4.2); it was a property of the reference field, not of the algorithm. What remains true, and is the point of the limitation, is the structural one: because `w_phi` freezes wavenumbers, whatever the initializer misses is unrecoverable thereafter, so tracking quality is capped by initialization quality regardless of why a component was missed. The net benefit of tracking is therefore concentrated in providing real-time computational efficiency ($\sim$10 gradient steps vs. $\sim$200 for full re-initialization) rather than accuracy improvement beyond initialization. The corrected tracker sharpens this into a quantitative statement. Comparing per-episode post-initialization against post-tracking nRMSE across the 24 arm–episode pairs of the re-measured ablation and model-order experiments, tracking ends *above* its own initialization on 20 of 24, but by a small and bounded margin (+0.001 to +0.022 nRMSE; median +0.003). So the honest characterisation of Phase 2 on this dataset is not that it degrades the estimate—the earlier 2–3$\times$ inflation and the $0.009 \rightarrow 0.08$ collapse on the calmest episodes were properties of the uncorrected tracker and do not reproduce—but that it holds the initialization roughly in place at a fraction of the per-frame cost. Tracking is a compute mechanism, not an accuracy mechanism. *Residual amplitude under-estimation*: the aggregate variance ratio is 0.916, i.e. $\sim$8% amplitude under-estimation, and it is regime-independent (VR = 0.910 for $\leq$5-component and 0.925 for $\geq$15-component episodes) even though nRMSE and ρ differ sharply across the same split. We previously attributed this deficit to Adam momentum accumulating oscillating gradients from an inaccurate phase prediction $\phi_i \mathrel{+}= \omega_i \Delta t$. That attribution is withdrawn: the specific inaccuracy it named was a double application of the phase advance, which has since been corrected, and the deficit narrowed from 0.66 on the original mesh through 0.83 to the present 0.916

without the momentum path being touched. The remaining 8% is therefore *unattributed*—we can bound it but not currently explain it, and we prefer to say so rather than carry forward a mechanism whose named cause no longer exists. Post-hoc least-squares amplitude calibration is effective when dense observations are available but does not improve with sparse LiDAR ($\sim$50 pts/frame) due to insufficient spatial coverage; isolating the residual mechanism (momentum-free optimizers for amplitude, periodic optimizer state resets, or a per-frame amplitude least-squares solve) is future work. *Initialization resolution and the peak-separation radius*: the greedy beamforming peak search enforces a fixed minimum separation $\Delta k_{\min} = 0.15$ rad/m between accepted peaks, so component pairs closer than this in wavenumber space are structurally unresolvable by our initializer regardless of signal-to-noise ratio. This is not a corner case: 126 of the 195 simulated episodes—and every 15-, 20- and 30-component episode—contain at least one true pair separated by less than 0.15 rad/m. A pre-registered sweep of $\Delta k_{\min}$ (5 seeds per cell, decision rule fixed in advance, with a control episode containing no close pair) shows the constant to be a regime-dependent trade rather than a universal floor: relaxing it to 0.05 rad/m improves the crowded 15-component case on 5/5 seeds by 0.015 nRMSE—5.2% of that regime's error—but degrades the 5-component case on 0/5 seeds (+0.029 nRMSE), because a smaller radius lets the greedy search stack duplicate peaks on a single true component; the control episode is unaffected ($t = 0.72$). Two consequences follow. First, the crowded-regime error is *not* explained away by this constant: the bulk of it survives when the constant is removed. Second, statements in this paper of the form "these components are not recovered" are properties of the estimator as configured, and understate what this estimator class can achieve in crowded seas by a few percent; they should not be read as method-independent impossibility. The principled repair is not a smaller fixed radius but an adaptive one, scaled to the beamforming main-lobe width and hence to the effective aperture; we have not tested it. For $N_{\text{true}} \geq 30$, wavenumber-space crowding additionally exceeds what the beamforming grid resolves at any separation radius; higher-resolution or 3D (k_x, k_y, ω) FFT could help. *Current estimation*: we do not

estimate ocean currents, whose Doppler shift makes the apparent dispersion $\omega = \sqrt{g|k|\tanh(|k|d)} + \mathbf{k}\cdot\mathbf{U}$; a 0.5 m/s current introduces ∼5% frequency bias at BF4. Joint current–wave estimation is a natural extension. *Cold start and re-initialization*: Phase 1 requires $T_{\text{init}} = 2T_p$ of observations (e.g., ∼16 s at BF4) before tracking begins; during this interval no wave estimate is available. If the sea state changes abruptly (e.g., vessel course change exposing a new swell system), re-initialization is needed, incurring the same latency. A background re-initialization scheme running in parallel with tracking could mitigate this in practice. *Surface normal approximation for real data*: in simulation we use the exact Gerstner surface normal for confidence weighting c_j, but real-data application must substitute the flat-surface approximation $\mathbf{n} \approx \hat{\mathbf{z}}$. The resulting bias scales with wave slope (up to ∼9% in $\cos\theta_j$ at grazing incidence at BF4); wave-tank validation is planned to characterise the end-to-end nRMSE impact. *Observation weight nomenclature*: the weighting $c_j = \cos\theta_j/(r_j/d_{\text{ref}})$ is a geometry-aware simplification motivated by the incidence-angle dependence of specular reflection and the inverse-square range attenuation. The exact Fresnel reflectance $R(\theta)$ [35] is not used directly, as it would additionally require surface-roughness and polarisation assumptions that are difficult to justify from observation alone. The name "geometry-aware" (previously "Fresnel-aware" in an earlier draft) more accurately reflects the functional form. *Baseline fairness*: the 24-configuration ETKF sweep (retracted corpus; archived) and the LETKF/GPR-disp comparison (retracted corpus; archived) show our conclusions hold under the best baseline configurations tested, though further tuning of LETKF (e.g., adaptive localization radius, inflation scheduling) or hybrid ensemble–spectral approaches remain unexplored. *Uncertainty quantification*: WaveSplat returns point estimates; Hessian-based confidence intervals or multi-start ensembles could provide uncertainty. *Directional spreading*: each component is monochromatic; representing continuous directional distributions (e.g., $\cos^{2s}$) would require directional or von Mises basis functions. *Ship motion*: the current simulation assumes a stationary sensor platform. In practice, heave, pitch, and roll of the vessel change the LiDAR beam pattern frame-to-frame;

an IMU/INS-based coordinate transform is a necessary preprocessing step for sea-going deployment, and the residual motion-compensation error constitutes an additional noise source not modelled here. *Near-surface false returns*: at 1550 nm the absorption length of sea water is approximately 1 mm [36], so essentially no light penetrates below the surface; sub-surface volume scattering is negligible. However, sea spray, whitecaps, and rain droplets above the nominal water surface can generate false LiDAR returns that are indistinguishable from surface elevation measurements. These super-surface returns introduce a positive elevation bias not captured by the additive noise model, and their prevalence increases with sea state; empirical calibration on real data is needed to assess their impact. *Relationship to physics-informed learning*: WaveSplat can be viewed as physics-constrained optimization with a hand-designed plane-wave basis, structurally analogous to PINNs that use a neural-network basis. The key advantage of the explicit plane-wave basis is threefold: (i) the dispersion relation is satisfied by construction rather than as a soft penalty, (ii) beamforming provides an efficient spectral initialization that neural bases lack, and (iii) the $4N$ parameter count is orders of magnitude smaller than typical PINN architectures, reducing overfitting risk with sparse observations. Hybrid approaches—e.g., learned initialization or learned observation weighting—could combine the interpretability of spectral components with the flexibility of neural representations.

7. Conclusion

We estimate phase-resolved ocean wave fields from sparse 3D point clouds as a small set of continuous-wavenumber spectral atoms under a dispersion constraint, with the initial configuration proposed by a learned set-valued prior. On crowded multi-component seas this lowers median normalised RMSE by -38.6% against the strongest eligible grid baseline we possess (62/64 episodes), and the result survives a non-linear reference field with out-of-corpus seeds (43/48) in a comparison where the baseline itself retains skill (47/48).

What we regard as the substantive contribution is not the margin but the *attribution*. Holding the deployed pipeline fixed and removing one property at a time, the learned prior accounts for -36.3% (45/48) — measured against an analytic package that itself clears a zero predictor on only 15/48 episodes and the continuity of the wavenumber for -38.6% (46/48, dose-responsive, and reproducing in the sparse stratum at 14/16). The gain also survives re-scoring inside the observation-density band our measured records occupy (45/48) and at a forecast rung that 11 of those 12 records cover (47/48) — though in the latter case the comparator degrades across the same interval, so the growing ratio must not be read as a growing advantage. Neither re-scoring was blind — the expected direction was known before the runs, so they are not blind replications — and that rung sits at the lower end of the measured lead band rather than spanning it. The two ablation magnitudes are *not additive* with each other or with the baseline margin — different corpora, different comparison arms — and we verified separately that the grid-snapping ablation is not confounded with the refit it necessarily performs.

What we did not obtain. On 12 fixed-station stereo records the learned prior does not reach the pre-registered threshold against a per-record zero-field reference under any arm configuration we tested, including one in which it is given the same median-of-k field ensemble that the deployed real-data path uses. That treatment does lower error by a confirmed margin and does move the endpoint, but not across the threshold. We state this as a measured boundary rather than as an omission: the representation that earns its gain on crowded synthetic seas has not been shown to earn it on the measured data we hold, and locating that boundary — rather than widening the margin — is where we believe the remaining work lies.

We also note two limits of the evidence itself. The synthetic gains are confined to the crowded stratum; against the deployed analytic package the sparse stratum does not separate. And the learned checkpoint was trained on the same distribution family as the synthetic evaluation, so "out of sample" on that corpus

means on the seed axis only.

CRediT authorship contribution statement

The sole author is responsible for all aspects of this work.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Code, simulation data, and experiment scripts will be deposited on a public repository upon acceptance. The complete 100-episode dataset can be regenerated from the released simulator with the provided configuration files and fixed random seeds, ensuring end-to-end reproducibility of the tables and figures that were computed on it

reported in this paper.

The buoy directional-spread statistics used for the external-validity stratification of the retracted corpus (archived) are third-party observations from the Coastal Data Information Program [37], retrieved over OPeNDAP on 2026-08-03. We release the station list, the pre-fixed sampling rule, the retrieval and estimator scripts, and the derived per-record spread values, so that the stratification is reproducible without redistributing CDIP data.

We could not locate a public, phase-resolved, two-dimensional *lidar* sea-surface dataset, and we state the scope of that search rather than assert non-existence. The measured evaluation in this paper therefore uses stereo reconstructions from fixed stations rather than lidar, which is itself a scope limit: the sensor that motivates the method is not the sensor the negative result was obtained with. We enumerated the collections of the USACE Field Research Facility THREDDS server to collection level—its `frf/` and `projects/` trees

and the nine collections at the server root—and found one candidate, a 30-file, three-day scanning-lidar wave-shape deployment; that campaign's lidar coverage is three days of a sixteen-day campaign, which is a property of the deployment rather than of our search. We also confirmed that no public download path exists for the processed free-ocean wave dataset (FOWD) product. Airborne scanning-lidar measurements of open-ocean waves do exist in the literature [12]; what we could not find is a public archive of the underlying point data. This is a statement about the reach of our search, and a dataset outside it would not surprise us.

Acknowledgments

The author thanks the Avikus engineering team for discussions on shipboard LiDAR integration and practical deployment considerations.

References

[1] O. M. Faltinsen, Hydrodynamics of High-Speed Marine Vehicles, Cambridge University Press, 2005.

[2] L. H. Holthuijsen, Waves in Oceanic and Coastal Waters, Cambridge University Press, 2007.

[3] N. Desmars, M. Hartmann, J. Behrendt, N. Hoffmann, M. Klein, Nonlinear deterministic reconstruction and prediction of remotely measured ocean surface waves, Journal of Fluid Mechanics 975 (2023) A8.

[4] R. Vettor, C. Guedes Soares, Development of a ship weather routing system, Ocean Engineering 123 (2016) 1–14.

[5] S. Park, Wave-preview feedforward control for hydrofoil vessels using LiDAR-based spectral wave estimation, Ocean Engineering (2026). Submitted.

[6] Det Norske Veritas, Recommended Practice DNV-RP-C205: Environmental Conditions and Environmental Loads, Technical Report, DNV, 2010.

[7] Datawell BV, Datawell Waverider Reference Manual, 2009.

[8] J. C. Nieto-Borge, G. R. Rodríguez, K. Hessner, P. I. González, Inversion of marine radar images for surface wave analysis, Journal of Atmospheric and Oceanic Technology 21 (2004) 1291–1300.

[9] I. R. Young, W. Rosenthal, F. Ziemer, A three-dimensional analysis of marine radar images for the determination of ocean wave directionality and surface currents, Journal of Geophysical Research: Oceans 90 (1985) 1049–1059.

[10] P. Queffeulou, Long-term validation of wave height measurements from altimeters, Marine Geodesy 27 (2004) 495–510.

[11] M. R. Belmont, J. M. K. Horwood, R. W. F. Thurley, J. Baker, Shallow angle wave profiling LIDAR, in: 2008 IEEE/OES 9th Working Conference on Current Measurement Technology, 2008, pp. 217–223. doi:10.1109/CCM.2008.4480871.

[12] P. A. Hwang, W. B. Krabill, W. Wright, R. N. Swift, E. J. Walsh, Airborne scanning lidar measurement of ocean waves, Remote Sensing of Environment 73 (2000) 236–246. doi:10.1016/S0034-4257(00)00098-5.

[13] N. Desmars, F. Bonnefoy, S. T. Grilli, G. Ducrozet, Y. Perignon, C.-A. Guérin, P. Ferrant, Experimental and numerical assessment of deterministic nonlinear ocean waves prediction algorithms using non-uniformly sampled wave gauges, Ocean Engineering 212 (2020) 107659. doi:10.1016/j.oceaneng.2020.107659.

[14] G. Evensen, The ensemble Kalman filter: Theoretical formulation and practical implementation, Ocean Dynamics 53 (2003) 343–367.

[15] B. Kerbl, G. Kopanas, T. Leimkühler, G. Drettakis, 3d Gaussian splatting for real-time radiance field rendering, in: ACM SIGGRAPH, 2023, pp. 1–14.

[16] M. A. Donelan, W. M. Drennan, A. K. Magnusson, Nonstationary analysis of the directional properties of propagating waves, Journal of Physical Oceanography 26 (1996) 1901–1914.

[17] F. Fucile, G. Bulian, C. Lugni, A probabilistic approach for the quantification of prediction error in deterministic phase-resolved wave forecasting, Ocean Engineering 163 (2018) 718–736. doi:10.1016/j.oceaneng.2018.04.079.

[18] P. Naaijen, K. Trulsen, E. Blondel-Couprie, Limits to the extent of the spatio-temporal domain for deterministic wave prediction, International Shipbuilding Progress 61 (2014) 203–223. doi:10.3233/ISP-140113.

[19] U. D. Nielsen, Estimations of on-site directional wave spectra from measured ship responses, Marine Structures 19 (2006) 33–69.

[20] B. R. Hunt, E. J. Kostelich, I. Szunyogh, Efficient data assimilation for spatiotemporal chaos: a local ensemble transform Kalman filter, Physica D: Nonlinear Phenomena 230 (2007) 112–126. doi:10.1016/j.physd.2006.11.008.

[21] H. L. Tolman, User manual and system documentation of WAVEWATCH III version 3.14, Technical Report 276, NOAA/NWS/NCEP/MMAB, 2009.

[22] F. Wang, Q. Zhu, C. Cai, X. Wang, R. Qiao, From points to waves: Fast ocean wave spatial–temporal fields estimation using ensemble transform Kalman filter with optical measurement, Coastal Engineering 197 (2025) 104690. doi:10.1016/j.coastaleng.2024.104690, closest competing work: sparse 3-D point assimilation into a grid state with an ETKF, on a vessel holding position. The paper also measures direct assimilation of

the sparse returns and reports limited improvement at the original sparse locations, and its outlook names a spectral-state formulation.

[23] I. R. Young, On the measurement of directional wave spectra, Applied Ocean Research 16 (1994) 283–294.

[24] J. Capon, High-resolution frequency-wavenumber spectrum analysis, Proceedings of the IEEE 57 (1969) 1408–1418.

[25] R. O. Schmidt, Multiple emitter location and signal parameter estimation, IEEE Transactions on Antennas and Propagation 34 (1986) 276–280.

[26] C. E. Rasmussen, C. K. I. Williams, Gaussian Processes for Machine Learning, MIT Press, 2006.

[27] M. K. Titsias, Variational learning of inducing variables in sparse Gaussian processes, in: Proceedings of the 12th International Conference on Artificial Intelligence and Statistics (AISTATS), 2009, pp. 567–574.

[28] M. Liu, S. Dong, J. Chen, Y. Zhang, Optimized quiescent period prediction under harsh sea states using a linear wave model based on physics-informed neural networks, Ocean Engineering 310 (2024) 118763. doi:10.1016/j.oceaneng.2024.118763.

[29] F. Mohaghegh, J. Murthy, M.-R. Alam, Rapid phase-resolved prediction of nonlinear dispersive waves using machine learning, Applied Ocean Research 117 (2021) 102920. doi:10.1016/j.apor.2021.102920.

[30] S. Ehlers, M. Klein, A. Heinlein, M. Wedler, N. Desmars, N. Hoffmann, M. Stender, Machine learning for phase-resolved reconstruction of nonlinear ocean wave surface elevations from sparse remote sensing data, Ocean Engineering 288 (2023) 116059. doi:10.1016/j.oceaneng.2023.116059.

[31] E. J. Candès, J. Romberg, T. Tao, Robust uncertainty principles: Exact signal reconstruction from highly incomplete frequency information, IEEE Transactions on Information Theory 52 (2006) 489–509.

[32] S. G. Mallat, Z. Zhang, Matching pursuits with time-frequency dictionaries, IEEE Transactions on Signal Processing 41 (1993) 3397–3415.

[33] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, in: Proceedings of the 3rd International Conference on Learning Representations (ICLR), 2015.

[34] K. Hasselmann, T. P. Barnett, E. Bouws, et al., Measurements of wind-wave growth and swell decay during the Joint North Sea Wave Project (JONSWAP), Ergänzungsheft zur Deutschen Hydrographischen Zeitschrift A8 (1973) 1–95.

[35] M. Born, E. Wolf, Principles of Optics: Electromagnetic Theory of Propagation, Interference and Diffraction of Light, 7th ed., Cambridge University Press, 1999.

[36] G. M. Hale, M. R. Querry, Optical constants of water in the 200-nm to 200-μm wavelength region, Applied Optics 12 (1973) 555–563.

[37] Coastal Data Information Program, CDIP wave observation archive, scripps institution of oceanography, university of california san diego, THREDDS/OPeNDAP archive, `https://thredds.cdip.ucsd.edu/thredds/catalog.html`, 2026. Directional moment variables `waveA1Value`, `waveB1Value` and `waveEnergyDensity`; 24 stations, 25,426 quality-flagged records accessed 2026-08-03.

Table 7: Raydrop model comparison on the Nyquist-adequate dataset: geometry-weighted (U) vs. unweighted (S) return sampling, seed-0 episodes. Δ%: relative nRMSE change from weighted to unweighted (negative = unweighted is better). *Provenance*: this diagnostic was measured on the earlier "Nyquist-adequate" corpus, i.e. the corpus whose headline comparison is retracted (the compared arms were never scored on the same frames). It is retained as a diagnostic of the estimator's own behaviour, which is a within-arm property and does not depend on that defect. The sensor-model robustness claim is now carried by Section 5.3. WaveSplat columns are the median over 3 estimator seeds (42/123/456) under the corrected tracker, re-measured 2026-08-03. ETKF uses the default configuration (n_{ens}=50) and is carried over from the earlier pass on the same dataset (the tracker correction affects only the WaveSplat arm); its nRMSE values are near the zero-predictor level and are variance-dominated (retracted corpus; archived), so the relative sensitivity (Δ%) is the quantity of interest, not the level—and that sensitivity is itself configuration-dependent, shrinking under an honestly re-tuned baseline (Section 5.8).

	WaveSplat			ETKF		
Episode	U	S	Δ%	U	S	Δ%
BF2 1c	0.063	**0.025**	−60.3	0.396	0.338	−14.6
BF3 1c	0.009	**0.008**	−4.4	0.444	0.378	−14.9
BF3 2c	**0.007**	0.009	+19.5	0.409	0.355	−13.1
BF3 5c	0.024	**0.019**	−21.9	0.441	0.359	−18.5
BF4 1c	0.004	**0.003**	−17.1	0.453	0.342	−24.4
BF4 5c	0.017	**0.012**	−32.7	0.394	0.367	−6.9
BF4 15c	0.272	**0.198**	−27.2	0.537	0.404	−24.7
BF4 30c	0.333	**0.269**	−19.2	0.495	0.441	−10.9
BF5 15c	**0.216**	0.247	+14.6	0.461	0.394	−14.6
BF5 30c	0.246	**0.220**	−10.6	0.516	0.442	−14.3
Avg	0.119	**0.101**	−15.2	0.454	0.382	−15.9

Table 8: Sparsity robustness: nRMSE at varying observation fractions. Average points ≈ 50 at 100%. *Provenance*: produced before the tracker phase-advance correction; pending re-run and not directly comparable with the corrected full-dataset figures of the retracted corpus (archived).

Episode	20%	40%	60%	80%	100%
BF3 2c	0.170	0.168	0.168	0.167	**0.052**
BF3 5c	0.204	0.194	0.195	0.193	**0.167**
BF4 5c	0.199	0.198	0.199	0.199	0.200
BF4 15c	0.302	**0.283**	0.294	0.278	0.302
BF5 15c	**0.274**	0.276	0.260	0.289	0.297

Table 9: Noise robustness: nRMSE under additive Gaussian noise $\mathcal{N}(0, \sigma^2)$ on LiDAR elevation observations. N_a: adaptively selected component count. Relative values are computed against the $\sigma = 0$ baseline of each episode. *Provenance*: this diagnostic was measured on the earlier "Nyquist-adequate" corpus, i.e. the corpus whose headline comparison is retracted (the compared arms were never scored on the same frames). It is retained as a diagnostic of the estimator's own behaviour, which is a within-arm property and does not depend on that defect. The noise-robustness claim is now scoped by Section 5.6.

Episode	σ=0	σ=0.02	σ=0.05	σ=0.10	$\Delta\%_{\max}$
BF3 5c	0.168 (N_a=4)	**0.149** (N_a=4)	0.160 (N_a=7)	0.190 (N_a=10)	+13.3
BF4 5c	0.200 (N_a=3)	0.200 (N_a=3)	0.201 (N_a=6)	**0.198** (N_a=9)	+0.7
BF4 15c	0.302 (N_a=19)	0.303 (N_a=19)	**0.276** (N_a=20)	0.294 (N_a=11)	+0.4
Avg	0.223	0.217	0.212	0.227	—

Table 10: Deterministic forecast accuracy: average metrics over 5 episodes (BF3 1c/2c, BF4 1c/5c/15c) at increasing forecast horizons Δt. Estimation uses $t \in [0, 20]$ s; forecast into $t \in [20, 30]$ s. *Provenance*: measured before the tracker phase-advance correction and not re-measured. That correction improved the nowcast nRMSE, so the levels here should be read as conservative; the horizon-invariance, which is the point of the table, is a within-row property and is unaffected. The levels of that improvement are not quoted — they come from the retracted corpus (§5.1 provenance note).

Δt	nRMSE	SS	ρ	VR
+0.5 s	0.218	+0.141	0.323	0.325
+1.0 s	0.218	+0.140	0.322	0.327
+2.0 s	0.219	+0.138	0.323	0.330
+5.0 s	0.218	+0.135	0.328	0.342
+10.0 s	0.219	+0.128	0.316	0.331